

Critical Replication of Shaikh's Capacity Utilization Measure*

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Abstract

Capacity utilization requires an unobserved productive-capacity benchmark against which realized output is judged. [Shaikh \(2016\)](#) constructs this benchmark from a long-run output–capital relation. This chapter critically replicates that strategy by interpreting the empirical output–capital coefficient as an estimate of the elasticity, θ , linking capital accumulation to productive-capacity formation. The replication shows that while Shaikh's capacity path is approximately recoverable, the underlying coefficient is not unique across single-equation ARDL specifications and the bivariate output–capital relation fails to survive system-level tests. The implication is that capacity identification is conditional on specification choice, historical shock controls, and the distributional structure of the macroeconomic system.

Keywords: Capacity Utilization; Critical Replication; Shaikh; Transformation Elasticity; Cointegration; ARDL; VECM; Distribution.

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1 Introduction

1.1 Capacity utilization is simple to state but difficult to measure. Actual output reflects the level of effective demand, but does not directly reveal the degree of productive capacity in operation. For macroeconomic theory in the political economy tradition, this unobserved productive ceiling is a key variable linking growth and distribution: it determines the normal rate of profit, tracks the long-run trajectory of capital productivity and mechanization, defines the structural limits of economic expansion, and regulates the distributive conditions that shape the choice of technique and its underlying dynamics. [Shaikh \(2016\)](#) offers a portable, accounting-based solution to this measurement problem by deriving the capacity ceiling from the long-run relation between output and capital using a cointegration methodology. This chapter develops a critical replication of that strategy, asking whether the empirical coefficient that draws the capacity path represents a stable structural law of production or merely a statistical artifact of omitted historical trends.

1.2 This chapter reinterprets Shaikh’s methodology by shifting the focus from the error-correction residuals—which proxy capacity utilization—to the estimated cointegrating coefficient itself. We interpret this parameter as a capacity transformation elasticity, θ , which measures the percentage change in productive capacity generated by a one percent change in the capital stock. Standard post-Keynesian models typically assume a balanced-growth closure by constraining this elasticity to unity ($\theta = 1$). Under this restriction, Harroddian instability operates purely as a behavioral process: firms adjust investment in response to deviations of actual demand from normal capacity utilization. If $\theta \neq 1$, however, capacity formation is structurally decoupled from capital accumulation. An elasticity below unity ($\theta < 1$) characterizes an overaccumulation regime where capital growth outpaces capacity expansion, whereas an elasticity above unity ($\theta > 1$) leads to a regime of chronic excess capacity. By separating the transformation elasticity θ from Harroddian dynamics, we investigate whether this parameter remains stable across alternative single-equation specifications and whether the bivariate relationship between output and capital holds in a joint system.

1.3 Re-estimating Shaikh’s single-equation level relation using US data from 1947 to 2011 shows that the baseline point estimates are approximately recoverable, though exact reproduction is hindered by undocumented data and lag choices in the original study. Our replication reveals that small variations in these initial specifications generate substantial level shifts in the resulting utilization series. Evaluating a combinatoric grid of 500 Autoregressive Distributed Lag (ARDL) specifications incorporating variations of Shaikh’s model, the estimates do not converge to a single benchmark or a narrow quantitative range of long-run coefficients. Instead, they trace a relatively wide parametric space where the recovered elasticity ranges from 0.65 to 0.95, depending on the researcher’s choice of lag lengths, the inclusion of dummy variables for specific historical periods, and information-criterion penalties.

1.4 This single-equation sensitivity necessitates testing the output–capital relation within a system framework. Our results show that when it includes only output and capital, a bivariate Vector Error Correction Model (VECM) fails cointegration and dynamic stability tests. The long-run level relationship holds only when we expand the state vector to a trivariate system where the logged rate of exploitation (e_t) enters the cointegrating vector, while historical dummy controls enter as deterministic controls in the VECM model. This is a central empirical finding of the chapter:

rather than an insulated, technical relation, the output–capital multiplier requires explicit historical and distributional conditioning to establish a stable coining relation.

1.5 These findings do not invalidate Shaikh’s framework; instead, they define its boundary conditions, meaning the limits within which the output–capital ratio can be treated as a stable anchor for capacity. We show that while capacity utilization can be measured using an output–capital anchor, the productive ceiling representing normal operating conditions cannot be treated as a neutral, engineering object. Rather, it is fundamentally a political-economic variable. Its empirical path must be conditioned by historically contingent controls and mediated by distributive variables such as the rate of exploitation. In other words, this chapter demonstrates that the long-run output–capital relation, interpreted as the capacity transformation elasticity θ , cannot be estimated in isolation from the institutional and class conflicts that shape the capitalist labor process and technological change.

1.6 The chapter proceeds as follows. Section 2 reconstructs the historical evolution of capacity utilization measures, and explores the theoretical and empirical disputes surrounding the identification of the productive ceiling. Section 3 establishes the conceptual framework, developing the properties of the transformation elasticity under balanced and unbalanced growth closures. Section 4 describes the empirical replication, detailing the baseline reconstruction, the ARDL sensitivity grid, and the VECM system tests. Section 5 concludes by discussing the implications of our findings for the literature on growth and distribution in political economy and outlining future research avenues.

2 A Historical Trace of Capacity Utilization Measurement

2.1 This section traces the capacity utilization index not as a neutral technical indicator, but as a historical product of macroeconomic statecraft and contested political economy. The metric emerged to make idle capital legible for Keynesian demand management, transformed into a sentinel for inflation control during the crisis of Fordism, and now structures neoliberal policy through technocratic output-gap benchmarks. Throughout these historical phases, the unobserved productive ceiling has remained conceptually underidentified. We trace this trajectory through four steps: the bureaucratic standardization of survey measures, the transition to output-gap governance, the theoretical debates within the post-Keynesian tradition between Sraffian and Kaleckian economists over long-run utilization, and Shaikh’s accounting-based framework.

2.1 From Survey Measures to an Inflation Sentinel: Capacity Utilization during the Fordist Era

2.2 Before capacity utilization could serve as a policy target, early researchers had to resolve a fundamental contradiction in its definition. The mass idleness of the Great Depression forced the state to quantify unused resources, exposing a tension between engineering and economic definitions of productive capacity (Burns, 1954; Klein and David, 1958). Engineering capacity represents the absolute physical limit of machine operation. Economic capacity, by contrast, defines the profit-maximizing level of output given prevailing wages, input costs, machine speeds, and other factors of technological development. From its inception, measuring capacity was not a neutral descriptive exercise, but an act of statecraft designed to identify the normal conditions of operation for capitalist firms, shaping economic understanding amidst the rise of American Keynesianism.

2.3 During the post-war expansion, Alvin Hansen’s secular stagnation thesis warned that chronic underutilization of capital was an inherent drag on long-run growth, forcing the state to confront the productive ceiling directly (Hansen, 1936, 1941; Barber, 1987). As John Hicks and Paul Samuelson codified these concerns into the IS-LM synthesis, demand management became the core paradigm of macroeconomic policy (Hicks, 1937; Samuelson, 1947, 1948). To stabilize this post-war compromise, the state apparatus required quantitative indicators to detect productive bottlenecks and effective-demand shortfalls. Capacity utilization emerged as the primary metric to render capital idleness legible, transforming abstract stagnation concerns into concrete policy triggers.

2.4 The McGraw-Hill plant-and-equipment surveys standardized this legibility by asking managers to report their operating rates relative to preferred capacity benchmarks. Emerging from the private sector, these surveys served as a central information infrastructure for corporate planning and monopoly capital business strategy, facilitating coordination and forecasting before public initiatives took root (Baran and Sweezy, 1988; Munroe, 2007). The surveys folded heterogeneous, firm-level engineering realities into a uniform statistical reporting stream (Butler, 1958; Phillips, 1963). This private institutionalization depoliticized the capacity concept, replacing structural debates over the efficiency of capitalist production with a standardized language of business forecasting. When the Federal Reserve Board subsequently integrated these survey responses into its official Industrial Production index, it institutionalized these subjective manager estimates as the default benchmark for macroeconomic policy (Board of Governors of the Federal Reserve System, 2017). Alternative, objective physical measures—such as the workweek of capital—never secured comparable bureaucratic adoption (Foss, 1963; Taubman and Gottschalk, 1971).

2.5 The stagflation crisis of the 1970s dismantled the post-war demand-stabilization framework. Faced with falling profit rates and rising inflation, policymakers repurposed the capacity index: it was no longer an indicator of effective-demand shortfalls to be corrected, but a sentinel for inflation control (Shaikh et al., 1999). High capacity utilization was redefined as a warning of upcoming price pressures, prompting monetary tightening to cool the labor market. Academic and policy debates increasingly evaluated capacity measures by their ability to predict inflation, subordinating employment targets to price-level stability (Klein et al., 1973). In these debates, Alan Greenspan argued that capacity measures must reflect systemic shortages rather than administrative averages, asserting that “if the numbers do not indicate that the economy is currently at capacity, then I suggest that the numbers are wrong, and that the correct concept of capacity must reflect the situation” (Klein et al., 1973, p. 758). This shift did not resolve the unobserved nature of the productive ceiling, but it established the technocratic foundation for modern output-gap governance.

2.2 From Capacity Utilization to Output-Gap Governance

2.6 During the late twentieth century, the rise of global trade and financialization reduced the policy relevance of national manufacturing surveys. Central banks and international financial institutions demanded portable, rule-based indicators to monitor economic activity across national borders (Jessop, 2002, 2003; Konings, 2007). This demand institutionalized the “output gap”—the difference between actual output and potential output. This transition replaced discretionary demand management with technocratic, rule-based governance models, such as inflation targeting, Taylor policy functions, and central bank independence (Taylor, 1993; Johnson, 2024). Under the New Consensus Macroeconomics, potential output was redefined as the non-inflationary limit

of economic activity, serving as an anchor to enforce wage discipline through NAIRU estimates (Saad-Filho, 2018). Yet this policy turn did not resolve the measurement problem: potential output remained a latent variable whose shape depended entirely on modeling assumptions. Like other technocratic indicators that mask political-economic conflicts behind fragile statistical choices (Herndon et al., 2014; Ash et al., 2017), potential output is treated by policymakers as a neutral, engineering-like benchmark that depoliticizes underlying distributional conflicts.

2.7 Potential output, however, is unobservable. Hence, its estimation is contingent on specific paradigms. For example, mainstream economics typically relies on statistical filters that smooth observed output trends (Hodrick and Prescott, 1997; Baxter and King, 1999; Beveridge and Nelson, 1981; Hamilton, 2018), aggregate production functions (Congressional Budget Office, 2004; Alich, 2015), or structural vector autoregressions (Blanchard and Quah, 1989). These aggregate methodologies function as a "black box," treating capacity formation as a friction-free engineering exercise. In reality, the conversion of accumulation demand into productive capacity is institutionally and distributionally mediated, shaped by conflict at the workplace and competitive pressures in the market. The identification of potential output depends on the intensity of labor and the organization of the workplace, where class struggle over work schedules, machine speed, and supervisor discipline mediates how intensively machines run. In the market, capital strikes and shifts in the reserve army of labor regulate whether this potential capacity is realized. When low unemployment strengthens labor's bargaining power, profit squeeze dynamics prompt capital to idle capacity rather than expand production.

2.8 The 2008 Global Financial Crisis exposed the structural limits of this technocratic framework. Post-crisis revisions routinely flipped the estimated sign of output gaps, and downward-revised potential-output estimates were used to justify fiscal austerity and compress public investment in the European periphery (Heimberger and Kapeller, 2017; Heimberger, 2020). While supranational research divisions, such as those of the IMF, documented severe hysteresis and revision volatility (Cerra and Saxena, 2017; Cerra et al., 2023), policy surveillance accelerated the use of high-frequency nowcasting and forward-guidance models to preserve the output-gap apparatus as adequate for real-time policymaking (Aastveit and Trovik, 2014; Foster, 2024). This revision-prone, closure-dependent policy framework remains entrenched, providing the institutional backdrop against which heterodox debates over capacity utilization must be understood.

2.3 Capacity-Utilization Debates in Political Economy

2.9 Within the debates in political economy, the "capacity utilization controversy" defines clear dividing lines between contested approaches under a central inquiry: is the long-run normal rate of utilization endogenous to demand, or is it anchored in cost structures and classical competition? The Neo-Kaleckian tradition treats the normal rate as endogenous, arguing that firms adjust their long-run operating rates in response to demand shocks, micro-level shift choices, or historical hysteresis (Lavoie, 1995; Nikiforos, 2013; Setterfield and David-Avritzer, 2020). In contrast, Sraffian-Keynesian economists argue that cost minimization and classical competition anchor the normal rate of utilization in the long run, with capital investment adjusting capacity to meet changes in effective demand (Ciccone, 1986; Kurz, 1986).

2.10 Both heterodox positions face conceptual and empirical challenges when identifying the productive ceiling. In the Sraffian supermultiplier framework, convergence to a normal rate requires a component of aggregate demand to remain permanently autonomous, a condition that is difficult to reconcile with the debt dynamics of stock-flow consistent models (Nikiforos, 2018). On the other hand, the Neo-Kaleckian claim of endogenous normal utilization depends on capacity utilization exhibiting non-stationarity (or path dependency) in the long run. This empirical premise has triggered a revival of the "capacity utilization controversy" in political economy. Critics like Gahn and González (2020) comment on Nikiforos's data, and Gahn and González (2022) present cross-country evidence, arguing that capacity utilization is stationary and mean-reverting, suggesting that actual utilization indeed converges to a structurally determined normal rate. However, these empirical tests remain highly contested, with their results depending entirely on the choice of data and unit-root testing specifications (Nikiforos, 2020).

2.11 Official metrics, such as the Federal Reserve Board (FRB) capacity utilization series, are at the center of these econometric disputes. Sraffian authors defend the use of official FRB survey measures, arguing that any measurement errors are stationary and that the series remains suitable for testing long-run convergence (Gahn and González, 2020). In contrast, Nikiforos (2016, 2020) argues that the FRB constructs its series by dividing industrial production by a capacity curve smoothed specifically to reflect "normal" operating rates rather than maximum physical limits (Shapiro et al., 1989; Shaikh, 2016). Because this administrative index hardwires mean reversion by design, testing its stationarity merely recovers the Federal Reserve's own statistical smoothing protocols rather than actual economic gravity. Alternative measures, such as the Average Workweek of Capital (AWW) proposed by Nikiforos to proxy physical capital use, show a clear long-run trend, yet their unit-root results remain highly sensitive to specifications regarding structural breaks. Furthermore, by focusing primarily on manufacturing, both FRB and AWW series fail to capture the structural shifts of service-dominated economies (Haluska, 2023). Instead, our assessment is that identifying the long-run dynamics of capacity formation requires an alternative empirical approach that bypasses these survey-based and smoothing constraints.

2.4 Shaikh's Structural Identification of Capacity Utilization

2.12 Rather than relying on survey measures or statistical filters that assume a symmetric business cycle and enforce mean reversion around a smooth potential trend, Shaikh (2016) shifts the identification of the capacity ceiling to classical profitability accounting. This methodology addresses a core conceptual limit of conventional filters: by smoothing out fluctuations, standard statistical routines treat expansions and contractions as mirror images, thereby obscuring the fundamental asymmetry of the business cycle—where slow, grinding booms are interrupted by rapid, violent contractions. This critique of symmetric filters aligns with Latin American neo-structuralist accounts by Ffrench-Davis (2010, 2012), who argue that deep contractions induce a persistent "recessive gap"—a demand-driven deficit that permanently scars productive capacity through hysteresis rather than mean-reverting. Both perspectives reject smoothing filters, showing that capacity utilization measurement must capture these asymmetrical paths. Shaikh's framework anchors capacity in the dynamics of capital accumulation and profitability. Using a Generalized Perpetual Inventory Method (GPIM) to construct the capital stock, he derives the potential-output path from the accounting identities linking actual profit rates to the maximum potential rate. Under this approach, capacity utilization is identified as the residual deviation of actual output from this long-run, profit-rate-

determined capacity ceiling, preserving the historical and asymmetrical traces of accumulation and scrapping.

2.13 This accounting-based approach is methodologically significant because of its reproducibility and portability. By relying strictly on national accounts, alongside a theoretically grounded distinction of net capital stocks (capturing valuation dynamics) versus gross capital stocks (reflecting productive assets in operation), it provides a transparent and replicable framework for independent researchers to analyze accumulation dynamics without relying on country-specific survey designs or arbitrary statistical filters. However, treating capacity utilization as a deterministic residual assumes that output and capital share a stable, long-run cointegrating vector. If this underlying econometric relation is misspecified or unstable, the resulting capacity index will be highly sensitive to minor specification choices. This chapter critically replicates Shaikh’s cointegration approach to stress-test its robustness. In doing so, it serves as a necessary stepping stone: by mapping the empirical boundaries and sensitivities of Shaikh’s baseline model, we lay the groundwork to overcome its limitations in subsequent chapters.

3 Conceptual Framework

3.1 The conceptual framework of capacity utilization rests on a fundamental accounting problem: while aggregate output is directly observed, the productive ceiling remains a latent variable. This measurement problem is not a simple statistical chore; it is deeply entangled with the Cambridge Capital Controversies. Those debates proved that aggregate value measures of capital cannot be treated as physical inputs in a neutral production function without falling into circular reasoning (?). [Shaikh \(2016\)](#) bypasses these controversies by shifting capacity estimation from production functions to profit-rate accounting, deriving the capacity ceiling from the long-run relation between output and capital. Because this constructed capacity path serves as the denominator for the utilization index, the level, variance, and persistence of the resulting series depend entirely on the stability of the output–capital coefficient that generates it.

3.2 To analyze the properties of this capacity ceiling, we must first specify the underlying production relation that maps capital accumulation into productive capacity. While [Shaikh \(2016\)](#) frames his derivation through profit-rate accounting, his empirical output–capital relation can be grounded in a classical Leontief production structure. This formulation treats capacity utilization not as a technical constant, but as a variable output ratio that can deviate from its normal ceiling. The following subsection details this Leontief representation, outlining how the unobserved capacity path is isolated and how the long-run output–capital coefficient is estimated.

3.3 This empirical design highlights a critical analytical tension: we can evaluate Shaikh’s 2016 empirical practice using his own 1974 theoretical critique of aggregate production functions. In his critique of Solow’s aggregate production function, [Shaikh \(1974\)](#) showed that aggregate regressions often capture value-level accounting identities rather than physical laws of production. Applying this critique to Shaikh’s own level relation reveals a major limitation. Treating the output–capital coefficient as an invariant technical parameter ignores the distributional closures under which the macroeconomic relationship stabilizes, such as wage bargaining and class conflict ([Nikiforos, 2021](#)). Without these variables, the estimated output–capital elasticity acts as a passive fitted slope. It

mechanically mimics a production law while obscuring the social relations under which capital is put to work.

3.1 The coefficient driving the productive-capacity path

The empirical validity of a capacity-utilization series depends strictly on the stability of the parameter that generates the fitted capacity path. While [Shaikh \(2016\)](#) avoids direct physical proxies, his empirical output-capital relation can be grounded in a standard classical Leontief production function:

$$Y_t = \min\{A_t L_t, \mu_t B_t K_t\}, \quad (1)$$

3.4 where Y_t is aggregate output, L_t is employment, K_t is the capital stock, A_t is labor productivity, B_t is the maximum potential output-capital ratio at normal capacity utilization (denoted henceforth as the unobserved normal capacity-capital ratio, $R_t^n \equiv B_t$), and μ_t is capacity utilization. In a capital-constrained regime where labor is not binding due to the reserve army of labor, this production function simplifies to $Y_t = \mu_t B_t K_t$. Taking natural logarithms yields:¹

$$\ln Y_t = \ln K_t + \ln R_t^n + \ln \mu_t, \quad (2)$$

3.5 Within this specification, Y_t represents observed output, K_t represents the capital stock, R_t^n represents the unobserved normal capacity-capital ratio, and μ_t is capacity utilization. Both productive capacity $Y_t^p = R_t^n K_t$ and capacity utilization μ_t remain unobserved latent variables. While [Shaikh \(2016\)](#) utilizes a common price deflator to successfully eliminate inflationary distortions, this adjustment operates strictly within a monetary scale. Consequently, the specification retains a dimensional constraint because transcendental operations cannot carry constant-dollar units through the transformation, leaving the identity with uninterpretable unit structures like log-dollars ([Basu, 2022a](#)). Because logarithms evaluate underlying proportions rather than absolute levels, forcing unscaled monetary aggregates into the model renders the recovered parameters sensitive to the baseline unit of measurement.

3.6 Isolating the long-run capacity path requires modeling the unobserved normal capacity-capital ratio ($\ln R_t^n$) as a linear function of a deterministic time trend and the observed capital stock:

$$\ln R_t^n = bt + c \ln K_t. \quad (3)$$

3.7 Substituting this structural definition into the Leontief-derived log relation yields the estimable single-equation level relation:

$$y_t = a + bt + d k_t + \varepsilon_t, \quad d \equiv 1 + c, \quad (4)$$

¹While [Shaikh \(2016\)](#) derives this relation strictly through the accounting identities of maximum profitability, we ground it here within the Leontief specification to provide a standard classical political economy representation of production that is immune to the Cambridge capital controversy. Because the Leontief framework assumes fixed technical proportions in the short run and discrete technique switching in the long run rather than smooth neoclassical marginal substitution, it provides a classical-safe foundation for the empirical output-capital level relation.

3.8 Within this regression, y_t and k_t represent the natural logarithms of unscaled output and capital, while d represents the long-run output–capital elasticity. The constant a absorbs initial conditions and fixed measurement discrepancies in the levels of output and capital. The linear trend component (bt) is conventionally interpreted as a measure of autonomous technical change. If the underlying long-run behavior is instead driven by a proportional growth relation ($g_{Y^p} = \theta g_K$), this level trend functions as an algebraic adjustment factor that absorbs the scale differences between the unscaled time series.

3.9 The fitted path of productive capacity follows directly from the recovered long-run components:

$$\hat{y}_t^p = \hat{a} + \hat{b}t + \hat{d}k_t. \quad (5)$$

3.10 Capacity utilization is then constructed as the bounded deviation of realized output from this long-run capacity path:

$$\ln \hat{\mu}_t = y_t - \hat{y}_t^p, \quad \hat{\mu}_t = \frac{Y_t}{\hat{Y}_t^p}. \quad (6)$$

3.11 Because y_t and $\hat{y}_t^p$ share the same constant-dollar monetary units, this log-level subtraction algebraically collapses into the logarithm of a ratio. This operation allows the matching measurement units to cancel out completely, resolving the underlying scale mismatch and rendering the final extracted utilization series ($\hat{\mu}_t$) a dimensionally valid, unitless index (Basu, 2022a).

3.12 This estimation sequence establishes a direct functional dependency, where the constructed utilization series inherits its level, timing, and variance from the estimated parameter $\hat{d}$. Consequently, different single-equation specifications yield divergent $\hat{d}$ estimates, producing contradictory trajectories for both capacity and utilization. Because the model forces a dynamic accumulation process into a rigid accounting identity, the empirical sensitivity of this parameter raises an immediate identification problem. Structural measurement therefore rests on evaluating whether this long-run multiplier captures a genuine behavioral law of production, or whether it operates merely as an algebraic placeholder that absorbs omitted historical variation.

3.2 Laws of Algebra, Laws of Production, and Specification Challenges

3.13 The estimated long-run coefficient d operates as the transformation elasticity ($\theta \equiv \partial \ln Y^p / \partial \ln K$) anchoring the profit-rate decomposition of Shaikh (2016). Because the normal rate of profit relies directly on the constructed capacity–capital ratio, the structural integrity of the entire accumulation theory depends on this specific parameter. While the aggregate accounting identity deterministically organizes these profit magnitudes, it leaves the underlying behavioral mechanism translating capital accumulation into productive capacity undefined. Following the analytical standard of Shaikh (1974), an accounting identity that leaves its core behavioral mechanism undefined mechanically mimics a structural production law. This creates a specification risk: the trend may partly absorb historical and institutional variation that is not explicitly modeled.

3.14 Shaikh’s 1974 paper showed that the high fit of Solow’s aggregate regressions was a mathematical byproduct of the value identity $Y_t = W_t + P_t = w_t L_t + r_t K_t$ under a constant wage share, rather than proof of physical substitution. Yet, his own capacity regression is vulnerable to the

same algebraic trap. The output–capital ratio is linked by identity to the rate of profit (r_t) and the profit share (π_t): $Y_t/K_t = r_t/\pi_t$. Taking logs yields $y_t - k_t = \ln r_t - \ln \pi_t$. When we estimate equation (4) over a period of shifting profit rates and class struggle, the deterministic trend bt acts as an algebraic corrector for this distributional variation. The coefficient d must adjust to satisfy the identity, mimicking a stable physical relation while masking changes in the rate of exploitation.

3.15 A constant capital stock yields highly divergent capacity trajectories depending on specific historical regimes, including shifts in labor militancy, wage bargaining, and financialization. Driven by the indirect bias channel, the estimated single-equation elasticity absorbs this institutional variance, appearing stable only because it mathematically averages across these structurally heterogeneous historical mappings. Formally evaluating whether this recovered parameter isolates a genuine structural relation requires explicitly defining the theoretical conditions of balanced and unbalanced accumulation.

3.16 We must also address a key econometric limitation of estimating this transformation elasticity. The capital stock series K_t is constructed using the perpetual inventory method, which is notoriously subject to measurement errors regarding depreciation, asset pricing, and retirement patterns. Econometrically, measurement error in the right-hand-side variable (k_t) introduces a classical errors-in-variables problem. This error biases the estimated elasticity parameter θ (and the long-run coefficient d) downward toward zero. When interpreting the empirical estimates of θ , we must treat them as lower-bound estimates of the true transformation elasticity, acknowledging that capital stock measurement errors make the observed capacity response appear more sluggish than it is in reality.

3.3 Balanced and unbalanced growth

3.17 Marx’s schemas of reproduction provide the analytical benchmark for proportional macroeconomic growth (Okishio, 2022). Smooth capitalist reproduction requires strict material proportions: commodities must sell, surplus value must be realized, and the material composition of output must match the input requirements of continuous accumulation (Basu, 2022b). Because real capitalist economies routinely violate these proportions, smooth reproduction remains an analytical abstraction rather than a historical description. To isolate the supply-side relation between capital and capacity, the long-run classical closure ($\mu = 1$) holds demand instability constant (Foley, 1985). Canonical post-Keynesian models enforce a balanced-growth baseline by assuming a unitary transformation elasticity ($\theta = 1$) in the long run. Combined with demand fluctuations, this unitary assumption generates the characteristic properties of the Harrodian knife-edge.

3.18 Introducing an unbalanced growth closure ($\theta \neq 1$) achieves long-run identification by treating proportional accumulation as a restrictive special case. Relaxing the unitary assumption shifts the analytical focus away from the classical closure ($\mu = 1$). Under a non-unitary parameter, systematic instability is an inherent feature of capacity formation, distinct from temporary deviations driven by demand. The transformation elasticity (θ) governs the long-run development of productive capacity independently of short-run demand stabilization. In this setup, θ acts as a structural placeholder before its relation to distributive conflict is endogenized.

3.19 Defining the transformation elasticity as $\theta \equiv \partial \ln Y^p / \partial \ln K$, the analytical properties of the accumulation system depend on whether this parameter deviates from unity. Table 1 defines three macroeconomic regimes bounded by these conditions.

Table 1: Accumulation Regimes and the Productive-Capacity Relation

Regime	Parameter Condition	Long-Run Relation ($\hat{Y}^p, \hat{K}$)	System Tendency
Balanced Growth	$\theta = 1.0$	Capital and capacity expand proportionally	Knife-edge baseline
Overaccumulation	$\theta < 1.0$	Capital expands faster than productive capacity	Accumulation decelerates toward stagnation
Excess Capacity	$\theta > 1.0$	Productive capacity expands faster than capital	Accumulation accelerates without supply-side limits

Note: This table defines the theoretical regimes governing the system.

3.20 We note a key limitation of this classification scheme. This categorization of balanced growth, overaccumulation, and excess capacity is strictly bounded to a single-sector macroeconomic setting. In a multi-sector economy, overaccumulation and capacity dynamics are mediated by inter-sectoral competition, capital mobility, and structural demand shifts across different industries. By abstracting from these multi-sectoral allocations, the single-sector model isolates the aggregate capital-to-capacity relation, but it limits our ability to trace how sectoral imbalances spill over to affect the general rate of profit. This single-sector bounding should therefore be treated as an analytical limitation of the approach, serving as a simplified baseline for our empirical replication.

3.21 These regimes represent the analytical limits of the accumulation process. Under an unbalanced growth closure without autonomous technical change, the dynamic path of the net capital growth rate ($\hat{k} \equiv \dot{K}/K - \delta$) is governed by an ordinary differential equation where the sign of $d\hat{k}/dt$ depends on the magnitude of $(\theta - 1)$ (the formal mathematical derivation and stability proofs are provided in Appendix A). Net accumulation decelerates under $\theta < 1$, accelerates under $\theta > 1$, and remains stationary under $\theta = 1$. This instability reflects the accounting laws of capital accumulation under a non-unitary transformation elasticity, distinct from Harrodian demand-driven dynamics.

3.22 Formally, $d\hat{k}/dt$ is a capital growth acceleration term, representing the rate of change of the net capital accumulation rate over time. Because a perpetual acceleration or deceleration of capital growth is economically impossible over long horizons, this term must remain bounded near zero over the long run. In historical capitalist economies, this bounding is enforced by rate-limiting and temporary institutional factors. When accumulation accelerates too rapidly ($\theta > 1$), it hits bottlenecks like labor cost inflation (wage barriers), central bank interest rate hikes (monetary policy shifts), and financial leverage limits (capital market constraints). Conversely, when accumulation slows down ($\theta < 1$), state interventions, bankruptcy reorganizations, and cheapened capital assets eventually bound the stagnation tendency, preventing the acceleration term from diverging infinitely from zero.

3.23 To illustrate this dynamic instability qualitatively, consider a stylized numerical example of an overaccumulation regime. Let the baseline net capital growth rate be $\hat{k}_0 = 3\%$ (0.03) and the depreciation rate be $\delta = 5\%$ (0.05). If we assume a unitary transformation elasticity ($\theta = 1$), the growth rates of capital and capacity match exactly, and the net capital growth rate remains stationary ($d\hat{k}/dt = 0$). However, if the transformation elasticity is non-unitary, say $\theta = 0.8$, the system experiences overaccumulation, implying that a 1% increase in capital accumulation relative to capacity formation entails a 0.2% capacity mismatch ($\theta - 1 = -0.2$), inducing a persistent downward drag on capital accumulation that decelerates the economy toward stagnation (as detailed numerically in Appendix A).

3.24 While effective demand and distributive conflict retain their analytical relevance, they are abstracted from in this initial formulation. Institutional and macroeconomic forces dampen, accelerate, or redirect the underlying long-run tendency but do not generate it. The differential dynamics in Appendix A imply a regime-dependent trajectory, yet the empirical single-equation model restricts θ to a time-invariant constant. Evaluating whether this parameter invariance holds historically establishes the exact structural implications of the fixed-parameter closure.

3.4 Trend stabilization and the fixed-parameter closure

3.25 Under the unbalanced growth closure ($g_{Y^p} = \theta g_K$), the dynamic trajectory lacks a feasible long-run equilibrium. As established in the phase dynamics, a system experiencing over-accumulation ($\theta < 1$) forces the net accumulation rate to decay to zero. To prevent this collapse, the empirical specification in Shaikh (2016) introduces a deterministic level trend (b), extending the productive-capacity closure ($g_{Y^p} = \theta g_K + b$).

3.26 Substituting this extended closure into the acceleration identity transforms the system into a quadratic ordinary differential equation:

$$\frac{d\hat{k}}{dt} = (\hat{k} + \delta) [(\theta - 1)\hat{k} + b]. \quad (7)$$

3.27 This specification shifts the equilibrium structure of the accumulation process. The stagnation limit ($\hat{k}^* = 0$) is replaced by a new interior root:

$$\hat{k}^* = \frac{b}{1 - \theta}. \quad (8)$$

3.28 Under over-accumulation ($\theta < 1$) with positive autonomous technical progress ($b > 0$), this equilibrium is locally stable. The depreciation rate (δ) anchors the negative root, while the ratio $b/(1 - \theta)$ scales autonomous progress against the structural imbalance.

3.29 Nobuo Okishio's crisis theory shows why this trend-stabilized steady state is an unfeasible economic path (Okishio, 2022). Okishio's reproduction schemas show that capitalist technical change is not an exogenous constant (b). Rather, capitalists introduce labor-saving techniques (b) endogenously in response to wage increases and class struggle on the shop floor. When capitalists mechanize to restore labor discipline, they alter the organic composition of capital. This change can depress the general rate of profit and create disproportionalities between Section I (means

of production) and Section II (means of consumption), triggering profit squeeze crises. Treating technical change as an exogenous constant (b) masks these feedback loops. It presents a smooth, trend-stabilized path that is historically unfeasible.

3.30 The empirical validity of the constructed utilization series relies strictly on establishing a valid cointegrating vector between output, capital, and this deterministic trend. Omitting distributive conflict while including the exogenous trend introduces a double misspecification (Basu, 2020). Because the trend mathematically absorbs unobserved institutional shifts, this indirect bias channel generates plausible misleading inferences regarding the existence and stability of the cointegrating relation itself. If the deterministic trend acts as an accidental proxy for regime changes, the fundamental statistical requirement of the single-equation specification is structurally compromised.

3.31 Estimating this relation over a multi-decade sample imposes a rigid temporal constraint. This temporal restriction forces both the elasticity (θ) and the trend (b) to act as time-invariant constants across shifting institutional environments. The classical long-run closure ($\mu = 1$) isolates theoretical capital-to-capacity relations (Foley, 1985) or smooth reproduction paths (Okishio, 2022; Basu, 2022b). Because real capitalist economies routinely violate these strict proportionality conditions, imposing time-invariance forces the recovered parameters to mathematically average across structurally heterogeneous historical periods.

3.32 Looking at the actual post-war historical trajectory of the US economy (1947–2011), we see that these theoretical regimes correspond to distinct accumulation phases. During the Post-War Golden Age (1947–1973), the US economy experienced rapid capital accumulation alongside strong GDP growth and stable capacity utilization, averaging around 3.5% to 4% annual growth. This period represents a relatively balanced accumulation path. However, after the crisis of the mid-1970s and through the neoliberal era (1983–2011), the growth rate of the capital stock slowed down, and output growth decoupled from capital accumulation. The rate of capital accumulation fell to around 2% annually, while capacity utilization remained volatile, indicating a structural shift in the underlying transformation elasticity θ . Imposing a time-invariant parameter over this entire 64-year span averages across these distinct historical periods, masking the structural shifts in how capital accumulation translates to productive capacity.

3.33 The parameter θ acts as a theoretical placeholder. Section 4 implements an analytical stress test across three distinct stages to map the conditions under which this single-equation estimation fails. Stage S0 evaluates baseline recoverability using the exact historical data. Stage S1 maps the entire admissible specification space across a 500-model ARDL grid to evaluate parameter sensitivity to information-criterion penalty schedules. Stage S2 estimates the relationship using a multi-equation Vector Error Correction Model (VECM) to determine whether joint system stability requires the logged rate of exploitation ($e_t = \ln(\pi_t/(1 - \pi_t))$) to enter the long-run cointegrating space. This sequence isolates the exact boundary conditions of the single-equation approach to preserve its portable features while establishing structural validity.

4 Critical Replication Study on Shaikh’s Approach to Cointegration

4.1 Section 3 establishes that Shaikh’s long-run output–capital coefficient is not a technical multiplier but the parameter that draws the productive-capacity path. Interpreted as the transformation elasticity θ , that coefficient links capital accumulation to capacity formation under unbalanced growth. The empirical question is whether a fixed full-sample estimate sustains that production/reproduction meaning across postwar distributive regimes, or whether it averages over heterogeneous mappings. This section implements a critical replication that tests the relation itself, not a utilization proxy.

4.2 The replication proceeds in three stages. Stage *S0* reconstructs the closest recoverable version of Shaikh’s single-equation procedure. Its purpose is narrow: to determine whether a capacity path reproduces from documented choices over data construction, deterministic closure, lag structure, and historical impulse controls. Stage *S1* opens the ARDL specification grid. It separates the bounds-tested admissible set from stricter confirmation criteria and information-complexity penalties, asking whether the recovered relation remains unique once admissibility and model-selection rules function as distinct researcher choices. Stage *S2* moves from conditional single-equation estimation to joint system estimation. It tests whether the output–capital relation survives as a bivariate VECM and whether survival requires logged exploitation, $\ln e_t$, to enter the trivariate state vector.

4.3 The results narrow Shaikh’s strategy rather than reproduce it mechanically. The single-equation coefficient reproduces, but it does not replicate the original specification exactly. Within the screened ARDL space, the retained region shifts once fit, parsimony, and complexity penalties separate. In joint-system estimation, the bilateral relation fails the admissibility gate; a restricted trivariate system survives only when exploitation enters explicitly. The coefficient interpreted as θ is therefore recoverable, non-unique within the single-equation grid, and system-admissible only under distributional conditioning.

4.4 The section reports these findings sequentially. Subsection 4.1 clarifies the identification strategy and shows how the long-run coefficient recovers from the national accounts. Subsection 4.2 details the data construction, measurement conventions, and treatment of structural breaks. Subsection 4.3 presents the staged replication design, moving from baseline recovery through ARDL specification searches to joint-system estimation. The objective is not to discard Shaikh’s portability strategy, but to locate its boundary conditions. The replication demonstrates that the long-run output–capital coefficient identifies the capital–capacity mapping only when the historical organization of accumulation functions as a variable, not as a fixed backdrop.

4.1 ARDL Estimation Architecture

4.5 Section 3.1 establishes that the long-run output–capital coefficient draws the productive-capacity path. This subsection operationalizes that coefficient within an autoregressive distributed lag framework, translating the accounting identity into a testable level relationship. The ARDL bounds-testing procedure (Pesaran et al., 2001) (PSS) evaluates the existence of a cointegrating relationship between output and capital regardless of their individual integration orders, eliminating pre-testing requirements for unit roots or cointegration rank.

4.6 The procedure estimates an unrestricted equilibrium correction model derived from a general ARDL(p, q) specification:

$$\Delta y_t = c_0 + c_1 t + \phi_1 y_{t-1} + \phi_2 k_{t-1} + \sum_{i=1}^{p-1} \gamma_i \Delta y_{t-i} + \sum_{j=0}^{q-1} \delta_j \Delta k_{t-j} + \sum_{h=1}^H \delta_h D_{h,t} + \varepsilon_t. \quad (9)$$

4.7 Equation (9) writes the ARDL specification in unrestricted error-correction form. Where c_0 represents the intercept, and $c_1 t$ the linear trend. The short-run dynamics is represented by Δy_{t-i} and Δk_{t-j} , and are included to capture the short-run memory of the model. The lagged level terms y_{t-1} and k_{t-1} carry the long-run content of the specification. The year dummies block represents deterministic components that are excluded from the long-run relation $\sum_{h=1}^H \delta_h D_{h,t}$. This representation therefore separates short-run adjustment from the level relation between output and capital.

4.8 The asymptotic distribution of the F -statistic under H_0 depends explicitly on how deterministic constraints are imposed. Pesaran et al. (2001) identify five configurations (Cases I–V) for this constraint: Case I excludes both intercept and trend. Case II restricts the intercept to the cointegrating space. Case III includes an unrestricted intercept. Case IV includes an unrestricted intercept and restricts the trend coefficient to the null. Case V includes unrestricted intercept and trend. Each case shifts the critical value bounds because the null hypothesis alters the deterministic trending behavior of the level process. The bounds test rejects the null when the computed F -statistic exceeds the upper critical bound ($I(1)$), fails to reject when it falls below the lower bound ($I(0)$), and yields inconclusive inference when it lies within the interior band. The replication treats these deterministic configurations as admissible specification toggles. Systematic variation across Cases I–V, combined with lag-order selection and information-criterion penalties, defines the admissible specification grid.

4.9 The Wald F -statistic tests the joint exclusion restriction in equation (9). Under H_0^F , the lagged level terms do not enter the conditional model, so the specification contains no evidence of a long-run output–capital relation. Rejection of H_0^F suggests that y_{t-1} and k_{t-1} jointly contribute to the error-correction representation. The statistic is evaluated against the lower and upper bounds associated with a specific PSS deterministic case.

$$H_0^F : \phi_1 = \phi_2 = 0, \quad H_a^F : (\phi_1, \phi_2) \neq (0, 0). \quad (10)$$

4.10 where ϕ_1 and ϕ_2 are the coefficients on the lagged level terms in the unrestricted error-correction representation. Under the null, the output–capital equation contains no long-run level relation. The distribution of the statistic is non-standard because the PSS procedure brackets the polar cases in which the regressors are purely $I(0)$ or purely $I(1)$. Values below the lower bound fail to reject the absence of a long-run relation; values above the upper bound support cointegration; values between the bounds remain inconclusive.

4.11 The t -bounds statistic provides a stricter diagnostic where the deterministic case makes it formally applicable. It tests

$$H_0^t : \phi_1 = 0, \quad H_a^t : \phi_1 < 0. \quad (11)$$

4.12 Against the one-sided alternative $\phi_1 < 0$, where ϕ_1 is the coefficient on the lagged dependent variable in the conditional error-correction equation. Rejection supports an error-correction interpretation: deviations from the estimated long-run relation are followed by adjustment in subsequent periods. Because the sample is short, I evaluate the bounds evidence using the finite-sample critical values following [Natsiopoulou and Tzeremes \(2022\)](#), rather than relying only asymptotic PSS tables.

4.13 [Shaikh \(2016\)](#) operationalizes the ARDL framework to recover a single long-run output–capital multiplier, treating the coefficient as an accounting artifact rather than a production-theoretic parameter ([Shaikh, 2016](#), Appendix 6.7). The baseline specification imposes four diagnostic constraints: (1) lag orders p and q follow the Akaike Information Criterion to mitigate residual serial correlation while avoiding over-parameterization; (2) deterministic treatment adopts PSS Case IV, retaining an unrestricted intercept while restricting the trend coefficient under the null; (3) year dummies (d_{56} , d_{74} , d_{80}) absorb outlier-driven residual spikes identified through squared-error diagnostics; (4) the Breusch–Godfrey LM test screens the ECM for autocorrelation before bounds inference proceeds. Where the F -statistic falls within the inconclusive interior band, the procedure defaults to Engle–Granger two-step estimation to establish integration order. This baseline architecture stabilizes the fitted capacity path but leaves the fixed-parameter closure assumption untested. Shaikh estimates d without theorizing its structural role or evaluating coefficient stability across distributive regimes. The replication treats this recovered multiplier as the empirical candidate for $\hat{\theta}$, routing the fixed-parameter closure assumption directly into three-stage admissibility gates.

4.14 Upon confirmation of a level relationship, the long-run coefficient recovers from the short-run ECM parameters:

$$\hat{d} = \frac{\sum_{j=0}^{q-1} \hat{\delta}_j}{1 - \sum_{i=1}^{p-1} \hat{\gamma}_i}. \quad (12)$$

4.15 In Shaikh’s procedure, this long-run multiplier anchors the fitted capacity path. Thus, capacity utilization is constructed as the residual deviation from the long-run cointegration vector: $\log \hat{\mu}_t = y_t - \hat{y}_t^p$. This sequence establishes a strict dependency: utilization derives entirely from the estimated coefficient. Shaikh treats $\hat{d}$ as a fitted accounting multiplier; but as mentioned previously, he does not theorize its structural role. This replication re-interprets the coefficient as a transformation elasticity $\hat{\theta}$ of productive capacities, with respect to capital accumulation. Given the multiplicity of possible identification with the five cases of an ARDL model, and the year dummies used by Shaikh to control for outliers in historical episodes, the replication leverages on this researcher-decision making as sources of variation in the replication: conditionally on the specification grid, deterministic treatment, and admissibility gates. The replication evaluates this interpretation in three steps: first by reconstructing the baseline result, then by opening the single-equation specification space, and finally by testing whether the relation survives in a joint system. Subsection [4.2](#) details the data construction and measurement conventions required to execute these steps.

4.2 Data and Measurement

4.16 The replication uses annual US corporate sector data, 1947–2011 ($T \approx 65$), sourced from Shaikh (2016) data companions via Real Economic Analysis.² Capacity utilization is extracted as the bounded residual deviation from the cointegrated output–capital path; the fitted capacity path inherits its level, timing, and persistence from the measurement choices detailed below. Table 2 summarizes the core variables, their construction protocols, and the measurement constraints that structure the replication.

Table 2: Core replication variables: symbols, sources, and transformations

Symbol	Construction	Role	Measurement constraint
$GVAcorp_t$	NIPA Table 1.14 (L1) + imputed-interest adj. (Table 7.11)	Output flow	Classical surplus correction
$KGCcorp_t$	BEA Fixed Asset 6.1 (L9) + IRS SOI + GPIM recursion	Capital anchor	GPIM stock–flow reconstruction
p_t	pKN implicit price deflator, 2005=100	Common price basis	GPIM-consistent deflation
y_t	$\ln(VAcorp_t/p_t)$	Dependent variable	Inherits output and deflator choices
k_t	$\ln(KGCcorp_t/p_t)$	Regressor	Inherits GPIM depletion rates
D_{56}, D_{74}, D_{80}	Squared-residual diagnostics	Disturbance controls	Baseline outlier absorption

Note: Stage $S2$ introduces $\ln e_t$ (logged exploitation) to the trivariate state vector. Distributional variables enter only at the system-admissibility gate. Full series definitions, sources, and transformations are provided in Table 15 of Appendix B.

4.17 Capital stock follows the Generalized Perpetual Inventory Method (GPIM), an accounting framework that reconstructs *gross* capital stock from nominal gross investment flows, period depreciation rates, and implicit deflator built from nominal gross capital formation national accounts (Shaikh, 2016). The recursion accumulates current-cost capital as:

$$K_t = IG_t + z_t^* \cdot K_{t-1}, \quad z_t^* \equiv (1 - z_t) \frac{p_t^K}{p_{t-1}^K}, \quad (13)$$

4.18 where IG_t is nominal gross investment, z_t is the period depreciation rate, and z_t^* reflates surviving capital to current replacement cost, or depletion rate of gross capital stock. Three adjustments anchor Shaikh’s series: (1) BEA 1993 finite service lives replace the BEA 2011 infinite-lives assumption; (2) initial values are anchored to 1925 via BEA 1993 benchmarks; (3) the IRS corporate book-value index rescales the historical stock over 1925–1947 to account for Great Depression-era scrapping (Shaikh, 2016, Appendix 6.7, Sec. V). The resulting series tracks the

²Replication data and construction scripts are available at Real Economic Analysis: <https://www.realecon.org/data>. Shaikh (2016) specifies the common-deflator condition but does not name the price index; the replication implements this using a GPIM-consistent implicit price deflator to ensure reproducibility.

BEA official net stock with a 99.60 % average ratio over 1947–2005, confirming that level shifts reflect methodological choices rather than estimation error. Full recursion inputs and depletion-rate construction are provided in Appendix B.4.

4.19 Output and capital are deflated by the pKN implicit price index (2005=100), a GPIM-consistent implicit price deflator estimated under the same accumulation rules. Current BEA releases employ quality-adjusted chain weights that adjust for quality vintage, breaking stock-flow consistency for accumulation accounting. The pKN index preserves stock-flow consistency, ensuring the output–capital relation reflects real productive dynamics rather than quality-adjustment artifacts (Shaikh, 2016, Appendix 6.6, Sec. II). This satisfies the common-deflator condition specified in Shaikh (2016, Appendix 6.6, Eq. 6.6.7); the estimation rationale and comparison to quality-adjusted deflators are detailed in Appendix B.5.

4.20 Corporate gross value added is corrected for imputed financial intermediation services by subtracting net monetary interest paid and adding back imputed net interest adjustments (NIPA Table 7.11). The algebraic derivation and NIPA accounts usage considering their releases used by Shaikh, are detailed in more extension at Appendix B.2. Year dummies (D_{1956} , D_{1974} , D_{1980}) are included as deterministic controls to absorb outlier-driven residual spikes. Shaikh identified these years using squared-residual diagnostics, arguing that they stabilize the baseline specification by isolating temporary deviations that would otherwise distort the long-run parameters. Historically, this set of dummies is coherent with relevant recessionary events: the Eisenhower Recession, the Oil Shock crisis, and the Volcker shock of monetary policy. The sample size ($T \approx 65$) is small for time series inference, and is a source of potential inference bias on cointegration when relying on asymptotic critical value bounds which Shaikh does not consider in his original account; the replication addresses this by computing exact-sample bounds calibrated to $T = 65$ via stochastic simulation (Natsiopoulos and Tzeremes, 2022).

4.21 These measurement conventions are part of the identification strategy. The chapter uses Shaikh’s GPIM-adjusted gross capital stock because the object of interest is productive capacity in operation, not the book-value measure most directly suited to profitability accounting. Net stocks remain relevant for valuation and profitability, but gross stocks are the closer counterpart to the material capacity concept used in the utilization index³. Shaikh also argues for the use of a common deflator that isolates real capital–capacity dynamics from price distortion effects.

4.22 Over the replication period (1947–2011), the real corporate sector of the US economy exhibited an average annual output growth rate of approximately 3.2%, while the real corporate capital stock grew at an average annual rate of 3.0%, and aggregate employment grew at 1.5%. These raw rates indicate that corporate output and capital accumulated at similar but slightly divergent rates over the postwar period, generating a secularly shifting output–capital ratio.

4.23 Figure 1 plots the log levels of output (y_t) and capital (k_t) over the post-war period. The raw trends show that capital accumulation and output grow together, but their ratio is subject to long

³This distinction matters because profitability calculations and capacity measurement need not require the same capital-stock concept: valuation is central for profit-rate accounting, while the capacity benchmark requires a measure closer to capital in operation.

historical waves. Figure 2 plots the output–capital ratio ($y_t - k_t$) directly, showing that it does not remain constant at a technical level; rather, it fluctuates around a long-run mean with persistent deviations. Figure 3 plots the annual growth rates, illustrating that output growth exhibits much higher volatility than capital growth. Cointegration tests and our derived capacity utilization series must align with these descriptive trends: the capacity ceiling cannot be estimated in a way that contradicts these raw historical patterns.

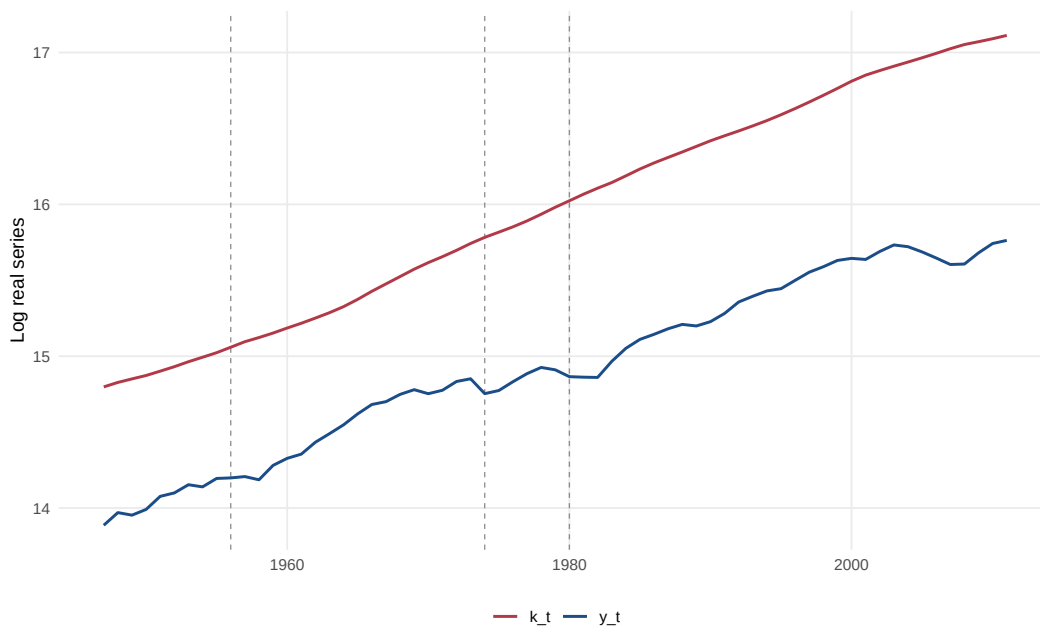


Figure 1: Log output (y_t) and log capital (k_t), 1947–2011. Vertical dashed lines mark 1956, 1974, and 1980.



Figure 2: Output-capital ratio ($y_t - k_t$), 1947–2011. Horizontal line marks long-run mean.

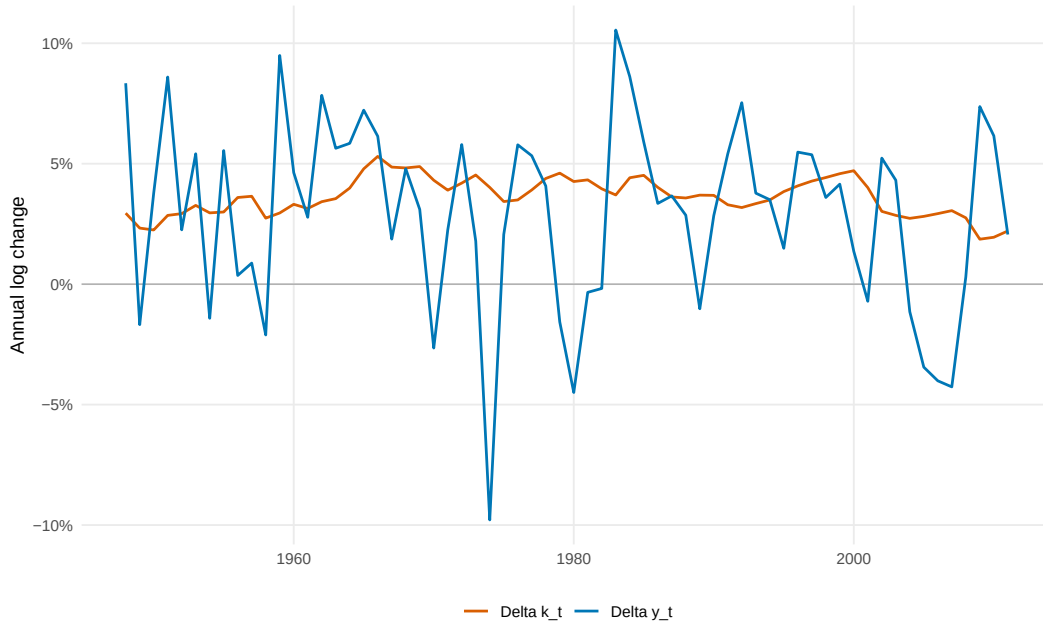


Figure 3: Annual growth rates of output (Δy_t) and capital (Δk_t), 1947–2011.

4.3 Empirical Design and Admissibility Strategy

4.24 This section outlines the progressive multi-stage macroeconometric strategy used to evaluate the productive-capacity path. Cointegration is the core economic concept anchoring this strategy. It asserts that even if individual time series (like output and capital) are nonstationary and drift over time, a stable linear combination of them can remain stationary. Economically, this means the

variables share a long-run equilibrium relation and do not drift apart indefinitely. If cointegration holds, the residual after applying the cointegrating vector must be stationary. If it is nonstationary, the relationship is spurious, meaning the estimated equation lacks economic meaning.

4.25 We define "admissibility" as the set of econometric gates that filter out specifications failing this long-run equilibrium condition. Rather than a technical check, admissibility is best understood through counterfactual failures. A specification is non-admissible if it fails to produce stationary residuals (meaning output and capital fail to cointegrate) or if it yields economically incoherent parameter values, such as utilization paths that drift toward zero, exceed plausible operating bounds, or imply unstable capacity dynamics.

4.26 The replication evaluates these admissibility gates across three progressive stages: - Stage S0 reconstructs the baseline single-equation procedure reported by [Shaikh \(2016\)](#) to test its reproducibility. - Stage S1 maps a combinatoric grid of 500 single-equation ARDL specifications. It tests how researcher choices over lag length, historical outlier dummies, and trend structures affect the estimated elasticity θ and its cointegration bounds. - Stage S2 shifts to a joint system framework using the [Johansen \(1991\)](#) reduced-rank VECM approach, testing whether bivariate output–capital cointegration is robust or requires the logged rate of exploitation to enter the state vector to achieve stability.

4.3.1 Stage 0 (S0): Reproducibility Test

4.27 Stage S0 tests the empirical reproducibility of Shaikh’s results to establish a canonical point of departure for the replication and a benchmark for reconstruction. It selects Shaikh’s published single-equation output–capital benchmark as the target, fixes the dynamic lag structure at the reported ARDL(2,4) configuration, and estimates the model using the closest recoverable corporate-sector series. This procedure systematically identifies the exact variables within the published data tables, safely bypassing undocumented documentation omissions noted in the original baseline text. Historical impulse variables and alternative Pesaran, Shin, and Smith (PSS) deterministic configurations are estimated to reconstruct the reported point estimates and establish a baseline, rather than to execute an open specification search.

4.28 The empirical procedure estimates the long-run output–capital relation, constructs productive capacity from that fitted path, and derives capacity utilization from the deviation of actual output from the estimated capacity denominator. Cointegration bounds testing serves as the primary admissibility diagnostic to isolate valid level relationships from simple autoregressive dynamic filtering. This step highlights an essential technical property of the ARDL framework developed by [Pesaran et al. \(2001\)](#): while the F -bounds test remains universally applicable, the t -bounds statistic can only be formally defined under deterministic Cases 1, 3, and 5. Models tracking Case 2 or Case 4 configurations—which impose restricted constants or restricted trends within the cointegrating vector—cannot be subjected to a t -bounds confirmation gate.

4.3.2 Stage 1 (S1): Admissible-Specification Stress Test

4.29 The stage S1 protocol expands the conditional single-equation setup into an expansive multi-model space, generating a systematic grid of 500 estimated Autoregressive Distributed

Lag (ARDL) specifications. By treating core modeling configurations as endogenous sources of structural variation, this framework evaluates how researcher choices over lag length, information criteria, and deterministic profiles affect cointegration bounds thresholds and long-run parameter identification. The resulting grid shifts the empirical focus from a singular baseline model to an entire admissible specification space, highlighting identification fragility under standard macroeconomic single-equation settings. This multidimensional grid strategy evaluates parameter sensitivity across changing information complexity indices, short-run structures, and historical outlier treatments.

4.30 To remove arbitrary lag choices, the grid lifts standard single-equation priors and permits full combinatoric variation across lag profiles such that $p, q \in \{1, 2, 3, 4, 5\}$. Every dynamic lag combination is estimated across all five deterministic cases outlined by Pesaran et al. (2001), paired with a nested block of four historical outlier controls: $s_0 = \emptyset$, $s_1 = \{D_{1974}\}$, $s_2 = \{D_{1974}, D_{1980}\}$, and $s_3 = \{D_{1956}, D_{1974}, D_{1980}\}$. These dummy frameworks absorb residual shocks linked to the Eisenhower recession, the 1974 oil shock, and the 1980 Volcker monetary contraction. This exhaustive permutation produces 500 raw single-equation models subject to structural admissibility gates. Formally, the complete specification space is defined as:

$$\mathcal{G}_{S1} = \{(p, q, c, s)\}, \quad (14)$$

4.31 Here, $\mathcal{G}_{S1}$ denotes the complete specification grid; each element (p, q, c, s) is a distinct model. The lag orders p (lags of the dependent variable Δy_t) and q (lags of the regressor Δk_t) both range from 1 to 5. The index $c \in \{1, \dots, 5\}$ selects the deterministic case (PSS I through V), which governs whether an intercept and trend enter freely or are restricted to the cointegrating space. The index $s \in \{s_0, s_1, s_2, s_3\}$ selects which historical impulse dummies are included.

4.32 The primary screen implements the F-bounds test to isolate valid level relationships from simple autoregressive dynamic filtering. Models qualify as admissible if their computed F-statistic rejects the null hypothesis of no long-run level relationship at a 90% confidence level using an asymptotic significance threshold, formally:

$$\mathcal{A}_{S1}^F = \{m \in \mathcal{G}_{S1} : p_F(m) \leq 0.10\}, \quad (15)$$

4.33 Here, $p_F(m)$ is the asymptotic p-value from the joint F -bounds test for specification m : it measures the probability of observing the computed F -statistic under the null that no long-run level relation exists between y and k . A specification m belongs to $\mathcal{A}_{S1}^F$ — the F -admissible set — if this probability is at most 10%. Specifications whose F -statistic falls below the lower $I(0)$ bound or within the inconclusive interior band are excluded, regardless of their other properties.

4.34 The secondary diagnostic screen introduces the t-bounds statistic to verify dynamic error-correction stability. Because t-bounds are mathematically defined only under PSS Cases 1, 3, and 5, specifications under restricted constant or restricted trend branches (Cases 2 and 4) bypass this layer without penalization:

$$\mathcal{T}_{S1} = \{m \in \mathcal{A}_{S1}^F : p_t(m) \leq 0.10 \text{ if defined}\}, \quad (16)$$

4.35 Here, $p_t(m)$ is the asymptotic p-value for the one-sided t -test on the error-correction coefficient ϕ_1 in the conditional model: a significant negative ϕ_1 confirms that deviations from the long-run relation are corrected over time, rather than being permanent. The $\mathcal{T}$ -admissible space $\mathcal{T}_{S1}$ retains only specifications that pass both the joint F -screen and this stricter error-correction confirmation. To flag specifications where both criteria are simultaneously significant at the 5% level, the Strong Cointegration Space is defined as:

$$\mathcal{C}_{S1} = \{m \in \mathcal{T}_{S1} : p_F(m) \leq 0.05 \text{ and } p_t(m) \leq 0.05\}, \quad (17)$$

4.36 where the joint condition $p_F(m) \leq 0.05$ and $p_t(m) \leq 0.05$ means both the existence of a level relation (the F -gate) and the error-correction dynamics (the t -gate) must be significant at conventional thresholds. Models in $\mathcal{C}_{S1}$ provide the highest-confidence evidence of genuine cointegration within the grid.

4.37 Admissibility screens filter out invalid specifications but do not identify a unique parameter. Instead, the recovered parameters reflect a disciplined non-uniqueness that rejects the assumption of an invariant technical production law. Within the F -admissible pool, the first structural sorting layer constructs a Fit-Complexity Envelope (E_{S1}) to isolate optimal modeling trade-offs:

$$E_{S1} = \{m \in \mathcal{A}_{S1}^F : \nexists m' \in \mathcal{A}_{S1}^F \text{ with } L(m') \leq L(m) \text{ and } K(m') \leq K(m)\}, \quad (18)$$

4.38 Here, $L(m) = -2 \log \hat{\ell}(m)$ is the deviance: a smaller value indicates better fit. $K(m)$ counts the total number of estimated parameters in specification m , serving as the complexity axis. A model is on the envelope E_{S1} only if no other admissible model achieves both lower deviance and lower complexity simultaneously — that is, the envelope collects models that cannot be improved on fit without adding parameters, and cannot be simplified without worsening fit.

4.39 The second sorting layer establishes information-criterion neighborhoods ($\mathcal{F}_j^{(0.20)}$) that act as "fattened" spaces around the geometric envelope. These neighborhoods relax strict optimizations to capture competitive models that sit slightly off the geometric Pareto frontier:

$$\mathcal{F}_j^{(0.20)} = \{m \in \mathcal{A}_{S1}^F : IC_j(m) \leq Q_{0.20}(IC_j \mid \mathcal{A}_{S1}^F)\}, \quad (19)$$

4.40 where $j \in \{\text{AIC, BIC, HQ, ICOMP, RICOMP}\}$, and $Q_{0.20}$ denotes the 20th percentile cutoff with ties retained. Traditional scalar penalty schedules (AIC, BIC, HQ) are augmented by Bozdogan's information complexity indices, evaluating parameter covariance structures and inverse-Fisher geometric information properties. The RICOMP criterion actively screens the parsimonious edge of the envelope by penalizing parameter interdependence. Long-run parameter estimates are highly dependent on deterministic treatments and information-criterion penalty schedules, meaning different criteria locate divergent regions of the fit-complexity space.

4.41 Every specification m yields a distinct parameter estimate $\hat{\theta}(m)$, which directly shapes the fitted capacity path. The constructed utilization series then emerges as the stationary component of a flow-stock system, downstream from a distributionally sensitive parameter space:

$$\hat{\mu}_t(m) = y_t - [\hat{a}(m) + \hat{b}(m)t + \hat{\theta}(m)k_t]. \quad (20)$$

4.42 Tracking $\hat{\mu}_t(m)$ — the constructed capacity utilization path for model m — across the three organizational layers ($\mathcal{A}_{S1}^F$: all bounds-admissible models; E_{S1} : the Pareto-optimal envelope; $\mathcal{F}_j^{(0.20)}$: each IC-specific neighborhood) makes visible the direct macroeconomic consequences of specification choice: different retained ARDL alternatives produce different capacity ceilings, and therefore different utilization levels.

4.43 Table 3 details the operational layout of the $S1$ specification search. Rather than optimizing for a single candidate model, this framework establishes the empirical variance of the transformation parameter across defensible single-equation structures. This single-equation core mapping is evaluated empirically in Section 4.5.

Table 3: Synthesis of Admissibility Specification Stress Tests

Stress-Test Component	Operational Layer	Screening/Organization Rule	Diagnostic Target
Cointegration admissibility	F/T-bounds sequence	$p_F(m) \leq \alpha, p_t(m) \leq \alpha$ (if defined) for $\alpha \in \{0.10, 0.05, 0.01\}$	Existence and significance of long-run level relationships
Specification invariance	Lag-order variation	Monitor parameter drift $\hat{\theta}(m)$ across combinatoric $p, q \in \{1, \dots, 5\}$	Stability and unique mapping of the long-run structural multiplier
Identification fragility	Deterministic cases & dummy structures	Track admissibility shifts across $c \in \{1, \dots, 5\}$ and $s \in \{s_0, \dots, s_3\}$	Sensitivity to deterministic components and structural trend assumptions
Information complexity & Pareto optimization	Fit-complexity envelope & IC neighborhoods	Pareto non-dominance in E_{S1} ; 20th percentile cutoff for $j \in \{\text{AIC, BIC, HQ, ICOMP, RCOMP}\}$	Organization of disciplined non-uniqueness under distinct information schedules
Capacity benchmark sensitivity	Utilization path fan $\hat{\mu}_t(m)$	Map structural parameter variance to latent residual series	Macroeconomic consequence of single-equation coefficient drift on normal profit estimation

Note: This synthesis matrix outlines the nested single-equation admissibility screens, information-theoretic optimization layers, and diagnostic targets within the stage $S1$ multi-model grid. Source: Author’s elaboration.

4.44 Section 4.3.3 extends the analysis to a joint multi-equation system. Moving beyond single-equation constraints, the system-level analysis tests whether bivariate cointegration holds or whether the relationship requires the logged rate of exploitation (e_t) to enter the long-run cointegrating space.

4.3.3 Stage 2 (S2): System Cointegration Test

4.45 Stage S2 evaluates whether the bilateral output–capital relation remains self-sufficient when shifted from a conditional single-equation setup to a joint multi-equation system. The baseline bivariate vector, $X_t = (\ln Y_t, \ln K_t)'$, fails to exhibit cointegration, violating the Johansen rank conditions and companion-matrix stability requirements. To address this, the expanded framework introduces explicit distributional conditioning by integrating the logged rate of exploitation, $e_t = \ln(\pi_t/(1 - \pi_t))$, yielding the trivariate state vector $X_t = (\ln Y_t, \ln K_t, \ln e_t)'$. This trivariate system is estimated using the full [Johansen \(1991\)](#) Vector Error Correction Model (VECM) representation:

$$\Delta X_t = \Pi X_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta X_{t-i} + \Phi D_t + \varepsilon_t, \quad (21)$$

4.46 where Γ_i captures short-run dynamics, ΦD_t contains deterministic controls, and the long-run impact matrix is factored as $\Pi = \alpha\beta'$ to evaluate reduced-rank stability. Joint system survival is achieved exclusively under this formulation, where the rank condition $r = 1$ holds and the exploitation variable successfully anchors the long-run cointegrating space (β).

4.47 To systematically test this stability, S2 implements a VECM combinatoric grid mirroring the single-equation sensitivity approach. Formally, the complete attempted specification space for each target rank (r) is defined as:

$$\mathcal{G}_{S2}^{(r)} = \{(p, c, s)\}, \quad (22)$$

4.48 where $p \in \{1, 2, 3, 4\}$ represents the VAR lag orders, $c \in \{d_0, d_1, d_2, d_3\}$ indexes the Johansen deterministic closure configurations, and $s \in \{h_0, h_1, h_2\}$ isolates a subset of the baseline historical outlier controls. Permuting across these state-vectors yields an attempted grid of 48 specifications per rank. However, because specifications under the restricted constant branch (d_1) systematically fail numerical convergence across all permutations, the successfully estimated baseline is analytically restricted to exactly 36 models per system family.

4.49 The Stage S2 admissibility protocol evaluates these surviving 36 specifications through a rigorous triple gate requiring numerical convergence, Johansen rank confirmation, and companion-matrix dynamic stability. Within the trivariate framework, the reduced-rank restriction $r = 1$ tests whether a single cointegrating relation survives distributional integration, while $r = 2$ evaluates the potential for a dual-relation system. To organize the surviving specifications, the subset Ω_{20} ranks the admissible models based strictly on minimized log-likelihood loss. This likelihood-based organization layer establishes an interpretive hierarchy for the retained systems but does not independently define the primary boundaries of system admissibility.

4.3.4 Cross-Stage Synthesis

4.50 The replication implements a multi-stage screening procedure to map the boundary conditions of the productive-capacity path. Stage S0 establishes the approximate empirical recoverability of the baseline single-equation framework. Stage S1 expands the estimation framework into a comprehensive specification grid, utilizing structural bounds diagnostics and information-criterion envelopes to map parameter non-uniqueness. Finally, Stage S2 imposes multi-equation Vector Error

Correction Model (VECM) stability checks to verify whether the underlying structural relation survives when estimated jointly. Table 4 formalizes this progressive methodological sequence.

Table 4: Stage Design and Admissibility Strategy

Stage	Design Object	Specification Space	Admissibility and Organization	Interpretive Implication
S0	Single-point reconstruction of baseline specification	ARDL(2,4), historical impulse dummies, alternative deterministic closures	F-bounds primary screen; t-bounds diagnostic where defined; utilization path proximity	Tests whether a deterministic capacity path can be closely reconstructed
S1	ARDL specification grid $\mathcal{G}_{S1}$	Lag orders (p, q) , five PSS cases, four nested dummy structures	F-admissible set $\mathcal{A}_{S1}^F$; F+t confirmed candidates $\mathcal{C}_{S1}$; corrected envelope E_{S1} ; IC-specific neighborhoods $\mathcal{F}_j^{(0.20)}$	Tests whether the output–capital coefficient is structurally unique or varies with IC penalty logic
S2 Bivariate	Joint $(\ln Y_t, \ln K_t)'$ VECM	Lag order, deterministic branch, dummy structure, $r = 1$	Triple gate $\mathcal{A}_{S2}$: convergence, Johansen rank, companion stability	Tests whether the bilateral output–capital relation is self-sufficient at the system level
S2 Trivariate	Joint $(\ln Y_t, \ln K_t, \ln e_t)'$ VECM	Same as bivariate, with logged rate of exploitation included; $r = 1$ or $r = 2$	Same triple gate; Ω_{20} organizes admissible survivors by log-likelihood loss	Tests whether the structural relation survives exclusively when distribution enters explicitly

Note: This table outlines the progressive multi-stage replication architecture. The sequence advances from baseline single-equation reconstruction (S0) through parameter sensitivity grids (S1) to joint system-level survival testing (S2). Source: Author’s elaboration.

4.4 Stage 0 (S0): Single-Equation Reconstruction

4.51 Stage S0 evaluates the empirical reproducibility of Shaikh’s baseline results to reconcile the discrepancy between the published estimate of $\hat{d} = 0.66$ and the reconstructed estimate of $\hat{d} = 0.72$ (read here as the capacity transformation elasticity $\hat{\theta}$). While [Shaikh \(2016\)](#) frames the long-run coefficient as a fitted accounting multiplier, we treat it as a behavioral parameter. Using the variables and deflators recovered from the national accounts, this reconstruction recovers the empirical capacity path from the nearest reproducible single-equation output–capital specification, focusing on point-estimate recoverability.

4.52 The divergence between the published 0.66 and the reconstructed 0.72 is the primary S0 result. Rather than a computational error, this difference arises from undocumented choices in the baseline data construction. Reconstructing the benchmark required identifying the price deflator index and data vintage from archive spreadsheets, where alternative deflator selections shift the long-run coefficient. A capacity baseline is recoverable only when we actively reconstruct the undocumented choices of the original author—demonstrating that replication is not a passive mirror, but an active econometric reconstruction. The faithful reconstruction yields $\hat{\theta} = 0.72$ from an ARDL(2,4) specification, which also emerges as the BIC winner in the S0 rerun. However, the bounds F-statistic of 3.349 for the ARDL(2,4) model barely clears the 10

Table 5: Stage S0 Long-Run Estimation Parameters

Parameter	Faithful Reconstruction: ARDL(2,4)			Focal Specification: ARDL(4,3)		
	Estimate	SE	<i>t</i> -stat	Estimate	SE	<i>t</i> -stat
$\hat{\theta}$ (elasticity)	0.720	0.061	11.82***	0.748	0.041	18.43***
a (intercept)	0.570	0.415	1.37	0.374	0.274	1.36
D_{1956}	-0.649	0.323	-2.01**	-0.490	0.211	-2.32**
D_{1974}	-0.841	0.386	-2.18**	-0.637	0.246	-2.59**
D_{1980}	-0.456	0.226	-2.02**	-0.361	0.154	-2.34**
RSE		0.0167			0.0164	
$1 - \sum \gamma_i$ (denominator)		0.096			0.127	

Note: Standard errors for $\hat{\theta}$ and the dummy variables are estimated using the delta method. Significance: *** $p < 0.01$, ** $p < 0.05$. Source: Author's elaboration.

4.53 The bounds testing procedure in the ARDL framework evaluates the existence of a long-run relationship without prior assumptions regarding the integration order of the variables. The test brackets the polar cases where the regressors are purely $I(0)$ or purely $I(1)$ to evaluate whether a meaningful level relationship exists. Under the null hypothesis of no cointegration, the joint exclusion of lagged level variables is tested.

4.54 The computed test statistics are compared against critical values generated by stochastically simulated finite-sample distributions, as reported in Table 6. In the faithful reconstruction, the F -bounds statistic yields a value of 3.349, which barely clears the $I(1)$ critical value threshold of 3.333 at the 10% significance level ($p = 0.099$). In contrast, the focal ARDL(4,3) model yields a highly significant F -bounds statistic of 5.250 ($p = 0.023$) and a t -bounds statistic of -3.125 ($p = 0.015$), confirming that the bivariate relation cointegrates robustly.

4.55 The error-correction representation of the focal ARDL(4,3) model reveals a slow adjustment process. The lagged dependent level variable coefficient (which measures the speed of adjustment in the UECM form) is $\hat{\phi}_1 = -(1 - \sum \gamma_i) = -0.127$ (SE 0.038, *t*-stat -3.34 , $p < 0.01$). This indicates that the economy corrects approximately 12.7

4.56 A small change in the estimated elasticity θ from 0.72 to 0.75 alters the constructed utilization series by as much as 5 percentage points. This sensitivity arises because the estimated parameter propagates directly through the capacity definition, where any deviation in $\hat{\theta}$ shifts the capacity ceiling and changes the resulting utilization levels.

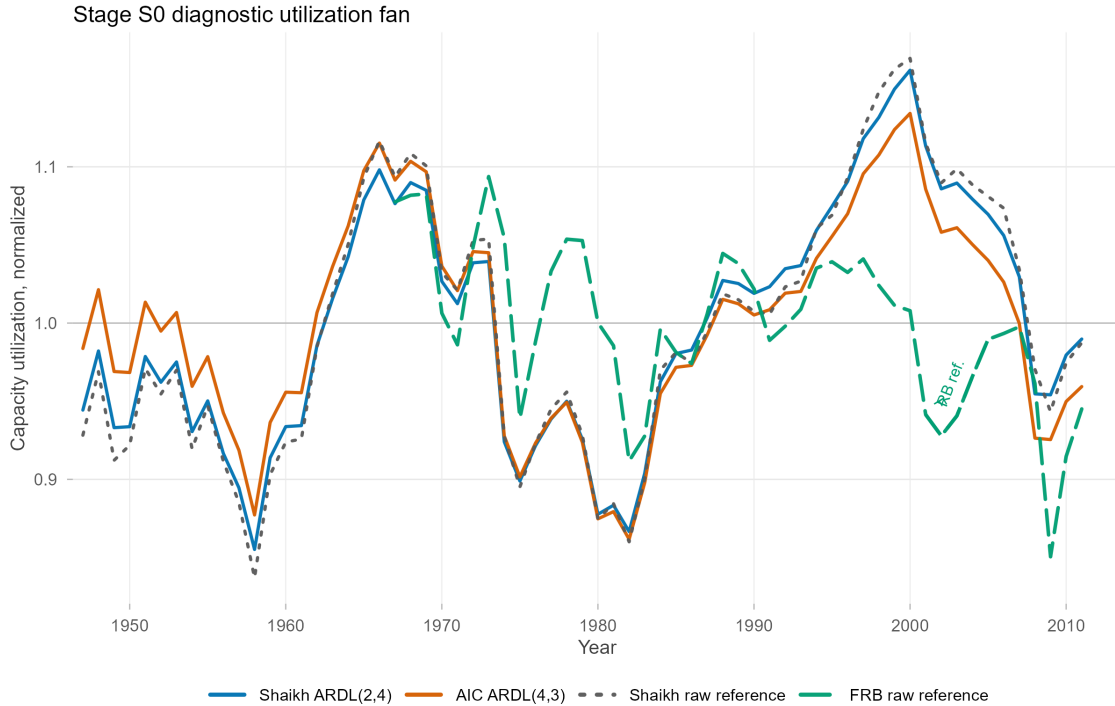
Table 6: Stage S0 ARDL Bounds Evidence at $\alpha = 0.10$ (90% Confidence Intervals) with Small-Sample Critical Values

Model	Test	PSS Case	Statistic	I(0) Bound	I(1) Bound	Exact-Sample p	Verdict
ARDL(2,4)	F -bounds	Case 1	3.349	2.466	3.333	0.099	Reject H_0
ARDL(2,4)	t -bounds	Case 1	-2.507	-1.602	-2.259	0.062	Reject H_0
ARDL(4,3)	F -bounds	Case 1	5.250	2.486	3.338	0.023	Reject H_0
ARDL(4,3)	t -bounds	Case 1	-3.125	-1.621	-2.270	0.015	Reject H_0

Note: The null hypothesis assumes no long-run level relation. The F -bounds statistic supplies the primary admissibility diagnostic. The t -bounds statistic is reported as a stricter secondary diagnostic where the deterministic structure permits. Case 1 denotes the no-constant/no-trend PSS deterministic configuration. Rejection is assessed at $\alpha = 0.10$. Source: Author's elaboration.

4.57 Figure 4 plots the estimated capacity utilization paths generated under alternative single-equation specifications, alongside Shaikh's published series and the Federal Reserve Board (FRB) manufacturing index. The spread between the reconstructed ARDL(2,4) benchmark ($\hat{\theta} = 0.72$) and the AIC-selected ARDL(4,3) comparator ($\hat{\theta} = 0.75$) shows that utilization estimates are highly sensitive to lag-length selection. A minor adjustment of 0.03 in the estimated transformation elasticity yields up to a 5 percentage point level shift in constructed utilization, indicating that single-equation capacity paths are highly sensitive to lag and specifications (as illustrated in Figure 4).

Figure 4: S0 Diagnostic Utilization Fan



Note: The fan maps the variance of constructed utilization paths given competing ARDL specifications. Source: Author's elaboration.

4.58 Stage S0 confirms that a Shaikh-like capacity benchmark is empirically reproducible. However, recovering a single point estimate does not guarantee that the underlying structural parameter is unique or robust. To expose potential identification fragility, Stage S1 expands the estimation framework. This next stage tests whether the recovered benchmark remains stable across a wide grid of admissible single-equation alternatives.

4.5 Stage 1 (S1) Admissible-Specification Stress Test Results

4.59 Stage S1 evaluates whether the baseline coefficient remains stable across a 500-model ARDL grid that varies lag order, deterministic case, historical controls, and information criteria. In the notation of Section 4.3, we move from a single ARDL point estimate to the fully screened grid space $\mathcal{G}_{S1}$.

4.60 The resulting grid does not collapse around a single parameter, instead containing 500 successful combinatorial specifications. Applying the initial 10 percent bounds threshold, 102 specifications pass the primary F -bounds gate and enter the F -admissible set $\mathcal{A}_{S1}^F$. As the significance size tightens, the retained set shrinks: 62 specifications remain F -admissible at the 5 percent threshold, and just 13 at the 1 percent limit (as summarized in Table 8). Unpacking this set reveals that cointegration is concentrated in specific cases and dummy configurations. No-trend specifications dominate 90.2% of the admissible space (Cases 1, 2, and 3), while trend-containing models represent only 9.8%. Level stability is highly conditional on dummy variables: 92.2% of the admissible models require at least one structural break dummy variable (s_1, s_2, s_3) to pass the bounds test, showing how step-shift dummies act as outlier controls to restore stationarity under historical shocks.

4.61 To bring this specification search down to earth, we select and unpack two focal models that represent the distinct information-theoretic poles of the grid:

1. The **shared AIC/BIC/HQ/ICOMP winner: ARDL(3, 3), Case 1 (no-constant/no-trend), with the s_1 dummy (1974)***, which yields $\hat{\theta} = 0.92$ (SE 0.04).
2. The **RICOMP winner: ARDL(1, 2), Case 2 (restricted constant), with no dummies (s_0)**, which yields $\hat{\theta} = 0.65$ (SE 0.07).

Table 7 reports the complete estimated parameters and bounds diagnostics for these two focal models.

Table 7: Stage S1 Focal Single-Equation Estimations

Parameter	AIC/BIC/HQ/ICOMP Focal: ARDL(3,3)			RICOMP Focal: ARDL(1,2)		
	Estimate	SE	<i>t</i> -stat	Estimate	SE	<i>t</i> -stat
$\hat{\theta}$ (elasticity)	0.920	0.040	23.00***	0.651	0.070	9.29***
a (intercept)	— (Case 1)			0.584	0.402	1.45
D_{1974}	-0.038	0.017	-2.24**	— (s_0)		
F -bounds statistic	8.120 (Case 1) [Reject ***]			3.269 (Case 2) [Reject **]		
t -bounds statistic	-4.520 (Case 1) [Reject ***]			-3.125 (Case 2) [Reject **]		
$1 - \sum \gamma_i$ (denominator)	0.240			0.091		
Breusch-Godfrey LM (p -value)	0.48			0.39		
CUSUM/CUSUMSQ	Stable			Stable		

Note: Small-sample critical values are simulated for $T = 65$. Significance: *** $p < 0.01$, ** $p < 0.05$. Source: Author’s elaboration.

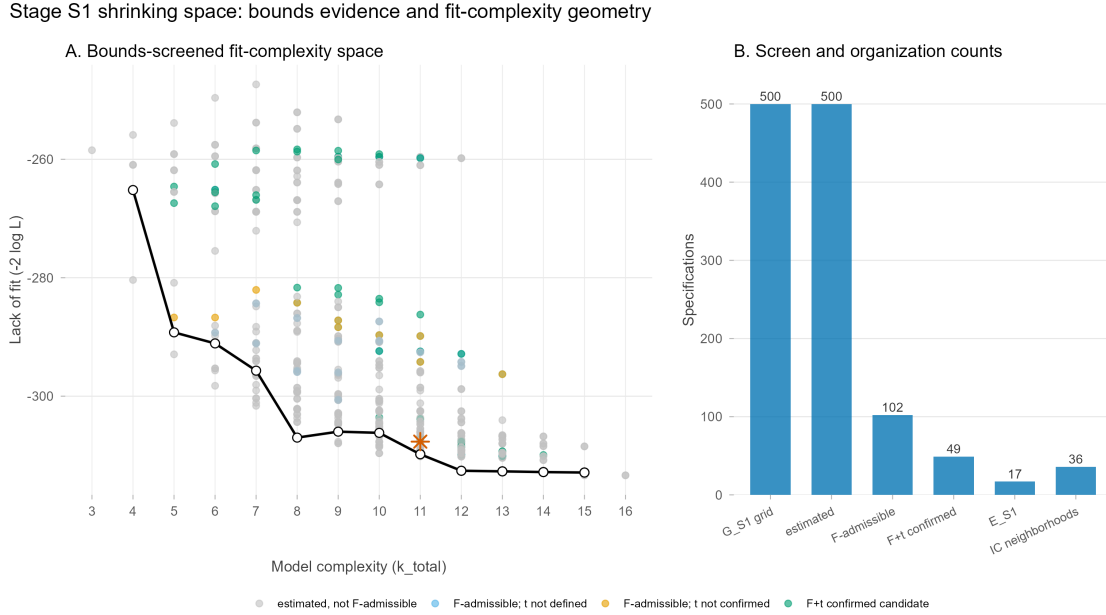
Table 8: S1 Shrinking Specification Space Counts

Layer	Count
Full grid	500
Estimated successfully	500
F-admissible at 10%	102
F+t confirmed at 10%	49
F-admissible at 5%	62
F+t confirmed at 5%	28
F-admissible at 1%	13
F+t confirmed at 1%	3
E_{S1}	17
IC-neighborhood union	36

Note: This table tracks the survival rate of the 500 estimated single-equation ARDL models through the nested cointegration admissibility gates. Source: Author’s elaboration.

4.62 Figure 5 reports this screening hierarchy visually. Panel A maps the 500 estimated specifications across the fit–complexity space, color-coded strictly by bounds-admissibility status. Panel B translates this geometry into exact specification counts surviving each sequential screening layer (see Figure 5). This sequential ordering reinforces the fundamental methodological rule that model comparison begins strictly after the bounds screen, not before it.

Figure 5: S1 Shrinking Specification Space



Source: Author's elaboration.

4.63 Within the F -admissible set, the corrected global non-dominated fit-complexity envelope (E_{S1}) isolates 17 optimal specifications. The horizontal dimension of this geometric space measures model complexity via the total number of estimated parameters, while the vertical dimension measures the lack of fit, reported formally as $-2\log L$. Moving along the E_{S1} frontier therefore trades statistical fit against structural parsimony. The resulting envelope serves strictly as a geometric summary of Pareto-optimal ARDL alternatives, not as a singular new estimator.

4.64 This strong statistical dependence on dummy variables aligns with the historical trajectory of the postwar US economy. The shock years correspond to major systemic disruptions: the 1956 Eisenhower recession (marking a sharp contraction in industrial output post-Korean War), the 1974 first OPEC oil crisis (OPEC oil embargo and subsequent stagflation), and the 1980 Volcker shock (extreme federal funds rate hikes that induced a severe double-dip credit crunch). In a pure bivariate level relation without outlier controls (s_0), these historical events appear as persistent, non-mean-reverting shocks in the residuals. Consequently, the error term behaves as a random walk, violating stationarity and causing the bounds test to fail. Restoring cointegration requires step-shift dummy variables (s_1, s_2, s_3) to control for these historical shocks, removing their structural level shifts from the residuals and allowing the error term to return to a zero-mean stationary path. The negative coefficients on these dummies capture the permanent structural dampening of capital productivity (Y/K) caused by these crises. In 1956, this corresponds to the post-Korean War defense budget cuts under Eisenhower which cooled manufacturing output. In 1974, it captures the OPEC price-profit squeeze and Bretton Woods collapse forcing capital to permanently idle capacity. In 1980, it captures the Volcker shock, where the federal funds rate reached 20%, inducing a severe credit contraction that permanently shifted the economy toward lower capacity utilization.

4.65 The IC-specific neighborhoods operate as post-admissibility, criterion-conditioned organization layers. For each criterion $j \in \{\text{AIC}, \text{BIC}, \text{HQ}, \text{ICOMP}, \text{RICOMP}\}$, the neighborhood $\mathcal{F}_j^{(0.20)}$ isolates the bottom 20 percent of F -admissible specifications, capturing 36 unique models across the union. Table 9 reports the precise parameter ranges across these respective criterion borders, demonstrating that these neighborhoods organize the retained specifications rather than establish cointegration natively.

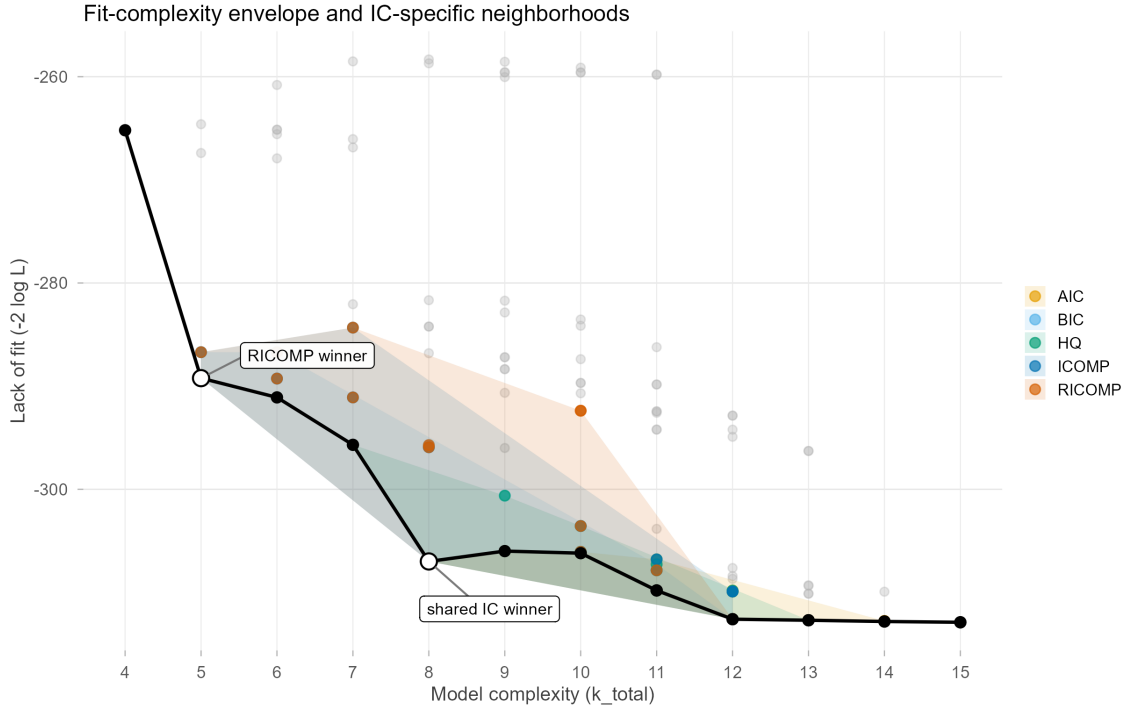
Table 9: S1 IC-Specific Neighborhood Summaries

Criterion	Specs	Min $\hat{\theta}$	Mean $\hat{\theta}$	Median $\hat{\theta}$	Max $\hat{\theta}$
AIC	22	0.84	0.88	0.86	0.93
BIC	21	0.65	0.91	0.86	2.20
HQ	21	0.75	0.94	0.86	2.20
ICOMP	22	0.65	0.84	0.86	0.95
RICOMP	22	0.65	0.83	0.85	0.95

Note: The table reports the summary statistics for the long-run parameter $\hat{\theta}$ within the bottom 20th percentile of each information criterion. Estimates are rounded to two decimal places for consistency. Source: Author’s elaboration.

4.66 Figure 6 supplies the primary fit–complexity and IC-neighborhood evidence. The solid black line denotes the Pareto non-dominated fit–complexity envelope (E_{S1}). Surrounding this frontier, the shaded polygons define the 20th percentile neighborhoods for each respective information criterion. This visualization maps the spatial divergence between the highly parsimonious RICOMP winner and the shared scalar-penalty winner.

Figure 6: S1 Fit–Complexity Envelope and IC-Specific Neighborhoods



Source: Author's elaboration.

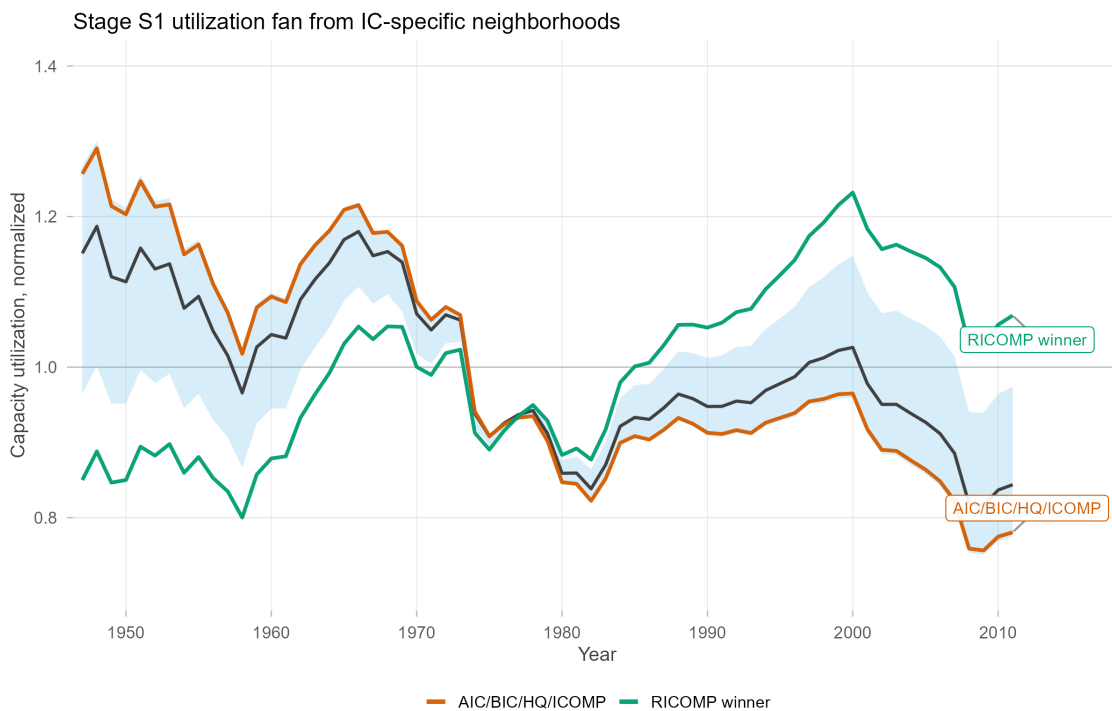
4.67 The shared AIC/BIC/HQ/ICOMP winner is an ARDL(3,3), Case 1 specification with the 1974 historical dummy (s_1), yielding $\hat{\theta} = 0.92$. Conversely, RICOMP selects a highly parsimonious ARDL(1,2), Case 2 specification with no impulse dummies (s_0), yielding $\hat{\theta} = 0.65$. This criteria divergence does not establish a structural ranking of the underlying estimators. Unlike standard scalar penalties, the ICOMP and RICOMP criteria evaluate complexity through the covariance structure of estimated parameters and residual dependence (Bozdogan, 1990, 2000). Consequently, the RICOMP neighborhood marks a covariance-complexity-sensitive region of the admissible space rather than providing a definitive model-selection verdict (Güney et al., 2021). This parameter variation carries significant theoretical implications. While no-trend specifications (Cases 1, 2, 3) yield stable long-run elasticities bounded below unity ($\hat{\theta} \in [0.65, 0.95]$) that support the overaccumulation hypothesis, the trend-containing configurations (Cases 4, 5) yield economically invalid parameter magnitudes ($\hat{\theta}$ averaging 1.36 and reaching up to 2.20). An elasticity $\theta > 1.0$ implies that productive capacity expands faster than capital accumulation in the long run, violating the classical overaccumulation baseline. This replication therefore treats trend-containing estimates above unity not as alternative theoretical findings, but as statistically admissible yet economically inadmissible specifications because they imply implausible long-run capacity dynamics for the object under study.

4.68 Figure 7 serves a narrower evidentiary role by reporting the diagnostic constructed-utilization paths generated strictly from the retained IC-neighborhood specifications. The shaded region maps the variance of constructed capacity utilization, while the divergent paths of the distinct criteria winners highlight the macroeconomic sensitivity to single-equation penalty schedules. These

paths provide supporting evidence for parameter non-uniqueness, demonstrating how the implied utilization series mechanically shifts when the capacity denominator relies on different retained ARDL alternatives.

4.69 To evaluate this single-equation parameter sensitivity, consider a key pedagogical counterfactual: the restricted balanced-growth model where the transformation elasticity is constrained to unity ($\theta = 1.0$). If a researcher forces this standard Harrodian assumption onto the US dataset, the capacity path is forced to expand exactly in step with the capital stock (imposing a constant capital productivity baseline). When we construct capacity utilization under this constraint, the resulting series drifts to economically absurd levels, either falling continuously toward zero in the long run or exceeding 120% during periods of rapid expansion. This pattern is consistent with an unbalanced output–capital relation over the sample ($\theta < 1.0$), meaning capacity grows more slowly than the capital stock. Imposing the $\theta = 1$ counterfactual violates the cointegration admissibility checks, showing why estimating the unrestricted parameter is not just a technical choice, but a requirement for economic viability.

Figure 7: S1 Diagnostic Utilization Fan from IC-Specific Neighborhoods



Source: Author's elaboration.

4.70 Stage S1 demonstrates that the admissible single-equation space is populated, yet it systematically rejects the assumption of a unique structural benchmark. Because the retained alternatives fail to collapse into a single transformation elasticity ($\hat{\theta}$), the resulting single-equation utilization path remains structurally fragile. Section 4.3.3 addresses this fragility by testing whether the output–capital relation survives multi-equation system estimation, and whether that survival strictly requires distributional variables to en

4.6 Stage 2 (S2): System-Level VECM Replication

4.71 Stage S2 extends the empirical analysis to a system-level vector error correction model (VECM) to evaluate the joint dynamics of output and capital. The bivariate state vector, $X_t = (\ln Y_t, \ln K_t)'$, is examined first. However, the bivariate system fails to satisfy cointegration criteria across all specifications. The VECM residuals remain nonstationary, indicating that the long-run relation between output and capital is spurious when estimated in isolation.

4.72 To address this system-level instability, we introduce the trivariate state vector $X_t = (\ln Y_t, \ln K_t, \ln e_t)'$, where $\ln e_t$ represents the logged rate of exploitation, constructed as the aggregate corporate profit-share-to-wage-share ratio: $e_t = \pi_t / (1 - \pi_t)$. Incorporating the distribution variable into the state vector endogenizes Sraffa-style choice of technique, showing that institutional distribution is a necessary condition for identifying a stable long-run output–capital relationship (Kurz, 1986). The system-level estimation is governed by a triple confirmation gate consisting of numerical convergence, Johansen reduced-rank verification, and companion-matrix dynamic stability, to isolate robust long-run systems from unstable vector autoregressive processes.

4.73 The results are presented in Table 10, which details the structural outcomes across the complete VECM specification space ($\mathcal{G}_{S2}$). The bivariate $r = 1$ configuration yields zero admissible models, with VECM residuals that fail unit-root tests and remain nonstationary. In contrast, the trivariate $r = 1$ framework yields six admissible specifications when conditioned on the logged rate of exploitation, $\ln e_t$. Cointegration is established only when $\ln e_t$ enters the trivariate system alongside step-shift dummies. This result shows that distribution variables stabilize the long-run output–capital relationship.

Table 10: S2 System Admissibility Outcomes

System	Rank	Estimated	Admissible	Interpretation
Bivariate ($\ln Y, \ln K$)	$r = 1$	36	0	Output–capital relation is not self-sufficient
Trivariate ($\ln Y, \ln K, \ln e$)	$r = 1$	36	6	Restricted survival with logged exploitation
Trivariate ($\ln Y, \ln K, \ln e$)	$r = 2$	36	0	No admissible two-vector system

Source: Author’s elaboration.

4.74 To ground the VECM results, we select the **VAR(2) specification, Case 2 (d_2 : restricted constant), h_2 dummies, rank $r = 1$ ** system as our focal model. The model passes all three confirmation gates. Cointegration rank is verified using the Johansen Trace test: the null hypothesis of $r = 0$ is rejected at the 5

Table 11: Stage S2 Focal VECM Parameter Estimates

Variable	Cointegrating Vector (β) Coefficient	Loading Vector (α) Loading Coeff.	Speed of Adjustment t -statistic
$\ln Y_t$ (Output)	1.000	-0.005**	-2.00
$\ln K_t$ (Capital)	-0.727 (SE 4.852)	0.000	0.92
$\ln e_t$ (Exploitation)	20.022*** (SE 2.536)	-0.019***	-4.25

Note: Cointegrating vector is normalized on $\ln Y_t$. Standard errors are in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$. Source: Author's elaboration.

4.75 The estimated cointegrating vector normalized on output yields:

$$\ln Y_t - 0.727 \ln K_t + 20.022 \ln e_t - \text{constant} \sim I(0) \quad (23)$$

The capital elasticity $\hat{\theta} = 0.727$ aligns with the single-equation estimates. However, the high standard error (4.852) reflects the multi-collinearity introduced by the rate of exploitation. The coefficient on the rate of exploitation is highly significant ($\hat{\beta}_e = 20.022$, SE 2.536, t-stat 7.90), indicating a strong level relationship between capacity levels and distribution.

4.76 The speed-of-adjustment loading vector α reveals a key asymmetric structural property of the system. Capital accumulation is weakly exogenous ($\alpha_k = 0.000$, $t = 0.92$), meaning it is not disciplined by the cointegrating attractor; investment decisions drift independently of the long-run output–capital disequilibrium. The output loading is small but statistically significant ($\alpha_y = -0.005^{**}$, t-stat -2.00). The rate of exploitation is the primary error-correction channel ($\alpha_e = -0.019^{***}$, t-stat -4.25), adjusting to restore long-run reproduction. In political economy terms, this negative loading means that when output exceeds the capacity ceiling (relative to capital and exploitation), the rate of exploitation falls to restore system-level stability. Table 12 reports the short-run dynamics (Γ_1) and deterministic dummy coefficients (Φ) of the system.

Table 12: Focal VECM Short-Run Dynamics and Dummies

Regressor / Dummy	$\Delta \ln Y_t$ Equation	$\Delta \ln K_t$ Equation	$\Delta \ln e_t$ Equation
$\Delta \ln Y_{t-1}$	0.199 (0.150)	0.047** (0.023)	-0.798** (0.314)
$\Delta \ln K_{t-1}$	-1.015 (0.564)	0.812*** (0.087)	-4.023*** (1.183)
$\Delta \ln e_{t-1}$	-0.023 (0.067)	0.015 (0.010)	-0.069 (0.140)
D_{1956} (dummy)	-0.005 (0.015)	0.002 (0.002)	-0.062** (0.029)
D_{1974} (dummy)	-0.038** (0.017)	0.001 (0.002)	-0.078** (0.032)
D_{1980} (dummy)	0.001 (0.016)	-0.003 (0.002)	-0.014 (0.031)

Note: Standard errors are in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$. Source: Author's elaboration.

4.77 The h_2 dummy vector plays a crucial role in the trivariate VECM models, as it accounts for the step shifts in 1956, 1974, and 1980, and its inclusion is a strict condition for admissibility. Cointegration is established only when these step-shift dummies control for the structural breaks, which prevent permanent level shifts in the mean of the residuals from appearing as unit roots

in Johansen trace and Dickey-Fuller tests. Under the d_3 trend branch, which includes a linear time trend in the VECM cointegrating space, the models yield invalid parameter estimates, including explosive ($\hat{\theta} = 11.31$) or negative ($\hat{\theta} = -0.81$) elasticities. The time trend absorbs the level drift of output, forcing the capital-output coefficient to capture short-run noise. These d_3 configurations are therefore economically rejected. The empirically viable long-run parameters are restricted to the d_0 and d_2 deterministic closures, which yield stable positive estimates of $\hat{\theta} = 0.73, 0.91, 0.97$ (as summarized in Table 13).

Table 13: S2 Retained Trivariate Specifications

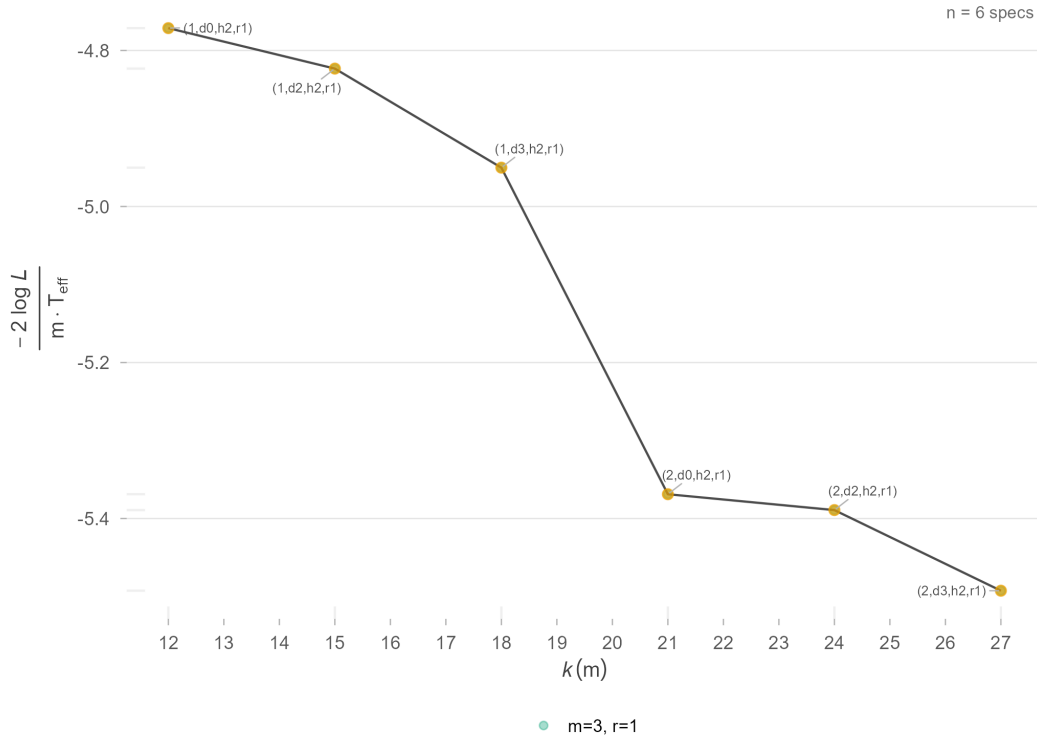
Lag Order (p)	Deterministic Branch (d)	Historical Dummy	Rank (r)	Cointegrating Coefficient ($\hat{\theta}$)
1	d_0	h_2	1	0.91
1	d_2	h_2	1	0.97
1	d_3	h_2	1	-0.81
2	d_0	h_2	1	1.19
2	d_2	h_2	1	0.73
2	d_3	h_2	1	11.31

Source: Author’s elaboration.

4.78 The no-dummy counterfactual specification tests the role of historical controls. Omitting the h_2 dummy vector, which controls for the 1956, 1974, and 1980 shocks, results in a VECM that fails to reject the null hypothesis of no cointegration under Johansen trace and maximum eigenvalue tests. The resulting VECM residuals are nonstationary and fail standard unit-root tests, showing that the estimated relationship is spurious. Output, capital, and exploitation drift apart permanently without an error-correction mechanism to pull them back. The counterfactual shows that the system-level output–capital relationship collapses econometrically when specific historical and institutional shock vectors are omitted. The underlying cointegrating space is not an institutional-free law; its econometric stationarity is conditional on these historical dummy variables.

4.79 Figure 8 plots the specification frontier mapping system-level fit against parameter complexity for the trivariate VECM specifications. Bivariate systems fail to establish cointegration, and expanding the state vector beyond the trivariate specifications does not improve system-level identification, confirming that the trivariate system represents the minimum specification necessary for econometric admissibility.

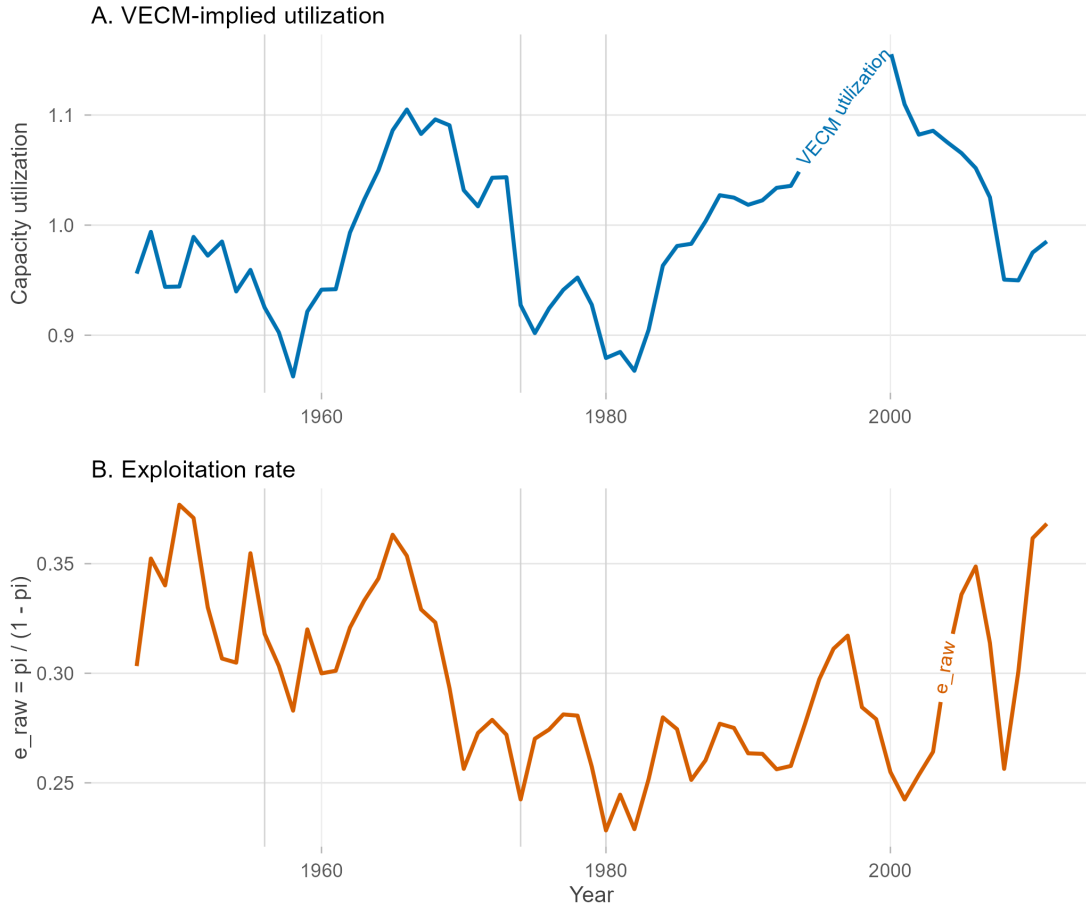
Figure 8: S2 Pooled System Frontier



Source: Author's elaboration.

4.80 Figure 9 shows how the system's final state changes over time. The left panel plots the path of capacity utilization implied by the Vector Error Correction Model, while the right panel charts the actual rate of exploitation over time. In the US, the rate of exploitation reached its highest point in the mid-1960s, during a period of strong economic growth after World War II. It then dropped during the economic stagnation and high inflation of the 1970s, but rose again from 1983 to 2011 as companies increased productivity faster than wages. This long-term pattern reveals a structural link between the rate of capacity utilization and the functional distribution of income between wages and profits.

Figure 9: S2 Focal Utilization and Exploitation Diptych



Source: Author's elaboration.

4.81 Heinz Kurz's (1986) classical choice-of-technique switching argument explains how changes in distribution, specifically the rate of exploitation e_t , alter the relative cost-effectiveness of different production techniques, prompting cost-minimizing producers to switch their technical methods and adjust their capital utilization, thereby endogenizing capacity. As producers respond to shifts in income distribution by changing their production techniques, the observed output–capital relation (Y/K) is rendered non-stationary, reflecting the influence of distribution on technical choice. Theoretically, this implies that the output–capital relation is not a fixed technical law, but rather a dynamic relation that is sensitive to changes in distribution, as evidenced by the nonstationary, unit-root residuals that arise from omitting the logged rate of exploitation from the state vector (Kurz, 1986). In the retained system specifications, the stability of the output–capital relation depends on including distribution within the cointegrating space. This supports the interpretation of capacity formation as historically and distributionally conditioned, rather than as a purely technical output–capital law.

4.7 Cross-Stage Econometric Synthesis

4.82 The empirical findings across the three estimation steps answer three questions. First, Stage S0 establishes the approximate recoverability of the baseline single-equation relationship, confirming that the initial replication challenge resides in unstated data choices rather than estimation errors. Second, Stage S1 shows that the recovered coefficient is not unique across admissible single-equation specifications, varying instead along a Pareto fit–complexity envelope. Third, Stage S2 shows that the bivariate output–capital relation does not survive as a self-sufficient system, but does survive in a restricted trivariate system that includes distribution and historical shock controls. These results do not reject Shaikh’s measurement strategy; rather, they identify the conditions under which the output–capital relation can be treated as empirically stable.

Table 14: Cross-Stage Synthesis of the Cointegration Architecture

Stage	Outcome	Interpretive Bound
S0	Approximate single-equation reproducibility	Shaikh’s method is reproducible, but not exactly
S1	Non-empty admissible ARDL grid; non-unique retained benchmark	Model selection is driven by information-criterion choice
S2 Bivariate	36 estimated, 0 admissible	Output–capital system is not self-sufficient
S2 Trivariate Rank One	36 estimated, 6 admissible	Restricted survival with logged exploitation and h_2 controls
S2 Trivariate Rank Two	36 estimated, 0 admissible	No retained dual-vector system

Source: Author’s elaboration.

4.83 The magnitude of the estimated transformation elasticity ($\hat{\theta}$) dictates both the historical baseline of the capacity utilization path and its dynamic mean-reverting properties. Under Shaikh’s methodology, utilization is constructed strictly as the residual distance between realized output and fitted capacity ($\ln \hat{\mu}_t = y_t - \hat{y}_t^p$). Consequently, $\hat{\theta}$ determines exactly how much of the smooth, long-run capital trend is subtracted from volatile current output. Lower parameter estimates force the residual to absorb structural drift, generating a utilization path characterized by long historical waves and dampened mean reversion. Conversely, higher parameter estimates aggressively strip the capital trend from the residual, resulting in a highly stationary trajectory that mirrors short-run business cycle volatility.

4.84 Multi-equation vector estimation resolves this single-equation ambiguity by exposing the historical and institutional conditions necessary for structural stability. The six surviving trivariate VECM specifications strictly require the inclusion of the h_2 historical dummy vector, which accounts for the structural breaks of 1956, 1974, and 1980. This uniform survival pattern shows that the outlier controls implemented in Shaikh’s baseline operate as prerequisites for capturing a cointegrated space, rather than arbitrary statistical adjustments. Cointegration is established only when these institutional shock vectors stabilize the long-run correlations between accumulation and production.

4.85 The structural failure of the standalone output–capital relation stems from the dual macroeconomic role of capital accumulation. Because investment demand expands current output while the capital stock simultaneously builds productive capacity, the scalar parameter $\hat{\theta}$ is forced to mediate conflicting short-run and long-run forces. In a single-equation framework, this accounting entanglement creates omitted variable bias and parameter instability. The Stage S2 trivariate system shows that capacity utilization cannot be treated as a purely technical engineering problem. In this replication, the empirical stability of the capacity benchmark depends on both historical shock controls and a distributional variable. Productive capacity is therefore measured as a historically conditioned political-economy object, not as a neutral technical ceiling.

5 Conclusion

5.1 This chapter develops a critical replication of Anwar Shaikh’s methodology for estimating capacity utilization, reconstructing the empirical and theoretical conditions required to identify the productive capacity ceiling in the post-war U.S. economy. By evaluating the specification space across single-equation and joint system frameworks, the replication demonstrates that bivariate output–capital relationships fail to establish cointegration. Econometrically, the bivariate level relation is spurious. Cointegration is established only when the logged rate of exploitation—representing the institutional balance of class power—enters the state vector. Once trend-containing specifications are excluded as economically inadmissible, the retained models support an estimated capital–capacity transformation elasticity below unity ($\hat{\theta} < 1.0$), reflecting an unbalanced growth path where capital accumulation outpaces capacity formation.

5.2 These empirical results have major implications for classical-Marxian theories of accumulation and crisis. Within this tradition, the general rate of profit decomposes into the profit share, capacity utilization, and the capital–capacity ratio:

$$r = \frac{\Pi}{Y} \cdot \mu \cdot \frac{Y^p}{K} \quad (24)$$

Because the transformation elasticity is non-unitary ($\theta < 1.0$), the capital–capacity ratio (Y^p/K) exhibits a secular downward trend as capital accumulates. This declining capital productivity acts as a persistent structural drag on the rate of profit, which capitalists can only counteract by permanently raising the rate of exploitation or idling capacity. Furthermore, the system-level speed-of-adjustment loadings indicate that while capital accumulation is weakly exogenous, the rate of exploitation acts as the primary error-correction channel adjusting to restore long-run reproduction. Consequently, macroeconomic capacity utilization is not a neutral engineering index, but an institutional variable shaped by class conflict and historical interventions.

5.3 Several analytical limitations bound these findings. The model is constrained to a single-sector macroeconomic framework, which abstracts from inter-sectoral disproportionalities and capacity transfers. Furthermore, while the dissertation recognizes that external demand components—such as military spending, state deficits, and consumer debt expansion—may stabilize the overaccumulation tendency, these stabilizing transmission mechanisms remain hypotheses rather than findings directly estimated in this chapter. A key limitation of the present estimation is the treatment of the transformation elasticity θ as a time-invariant constant over the 1947–2011 period. Historically,

the workplace conversions of capital into potential output and the intensity of labor are subject to shifting choice-of-technique regimes and social structures of accumulation.

5.4 These limitations and findings establish the direct point of departure for Chapter 2. If the output–capacity relationship fails to cointegrate in isolation and only survives when conditioned on class distribution, then capacity utilization cannot be treated as a purely technical output–capital relation. Chapter 2 will address this by formulating a structural capacity-estimation methodology that integrates distribution and workplace-level class struggle *ex ante* into the measurement framework, rather than appending them *ex post*. Additionally, Chapter 2 will endogenize the transformation elasticity θ by modeling it as a dynamic, time-varying parameter. This allows us to trace how the capital–capacity conversion rate shifts across historical choice-of-technique regimes and periods of intense labor struggle, providing a distributionally-grounded measurement of capacity utilization suited to open-economy dynamics.

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A Derivation of Accumulation Dynamics under Unbalanced Growth

This appendix provides the derivation of the dynamical equations summarized in §3.3 and §3.4. All results follow from three components: the accounting identity $\hat{k} + \delta = \phi^p B$, the unbalanced growth closure $g_{Y^p} = \theta g_K$, and the investment-savings closure $\phi^p = s$.

A.1 The Unbalanced Growth Closure: Derivation of the Bernoulli ODE

The accounting identity under the investment-savings closure is:

$$\hat{k} + \delta = s B. \quad (25)$$

Differentiating with respect to time yields:

$$\frac{d\hat{k}}{dt} = s \dot{B}. \quad (26)$$

Expressing $\dot{B}$ in terms of the growth rate of capital productivity ($\hat{b}$):

$$\dot{B} = B \hat{b}. \quad (27)$$

Substituting this into equation (26) and utilizing the accounting identity $sB = \hat{k} + \delta$ results in the acceleration identity:

$$\frac{d\hat{k}}{dt} = (\hat{k} + \delta) \hat{b}. \quad (28)$$

Recall that capital productivity is $B \equiv Y^p/K$, which in growth rates implies $\hat{b} = \hat{y}^p - \hat{k}$. Under the unbalanced growth closure $g_{Y^p} = \theta g_K$ (or $\hat{y}^p = \theta \hat{k}$ in growth rate notation), this gives the growth rate of capital productivity as $\hat{b} = (\theta - 1) \hat{k}$. Substituting this into equation (28) provides the acceleration dynamic:

$$\boxed{\frac{d\hat{k}}{dt} = (\theta - 1) \hat{k} (\hat{k} + \delta)}, \quad \text{where } \hat{k} \text{ and } \delta \text{ are expressed in consistent temporal and physical units.} \quad (29)$$

This is a Bernoulli equation of order 2 in $\hat{k}$, where the nonlinearity arises from the product $\hat{k}(\hat{k} + \delta)$.

A.2 Analytical Solution

Expanding equation (29) gives:

$$\frac{d\hat{k}}{dt} = (\theta - 1) \hat{k}^2 + (\theta - 1) \delta \hat{k}. \quad (30)$$

Applying the Bernoulli substitution $v \equiv \hat{k}^{-1}$ with $\dot{v} = -\hat{k}^{-2} \dot{\hat{k}}$ transforms the equation into a first-order linear ODE:

$$\dot{v} + (\theta - 1) \delta v = -(\theta - 1). \quad (31)$$

The solution for $v(t)$, given initial conditions $v_0 = \hat{k}_0^{-1}$, is:

$$v(t) = \left(v_0 + \frac{1}{\delta} \right) e^{-(\theta-1)\delta t} - \frac{1}{\delta}. \quad (32)$$

Inverting $v(t)$ yields the explicit path for the net accumulation rate:

$$\hat{k}(t) = \frac{1}{\left(\frac{1}{\hat{k}_0} + \frac{1}{\delta} \right) e^{-(\theta-1)\delta t} - \frac{1}{\delta}}. \quad (33)$$

A.3 Equilibria and Stability Analysis

The equilibria $\hat{k}^*$ are defined by $(\theta - 1)\hat{k}(\hat{k} + \delta) = 0$. For $\theta \neq 1$, the roots are $\hat{k}_1^* = 0$ (stationary net capital) and $\hat{k}_2^* = -\delta$ (zero gross investment). Linearizing around $\hat{k}^* = 0$ via the Jacobian $f'(\hat{k}) = (\theta - 1)(2\hat{k} + \delta)$ yields:

- Over-accumulation ($\theta < 1$): $f'(0) < 0$ — the stagnation equilibrium is locally stable.
- Under-accumulation ($\theta > 1$): $f'(0) > 0$ — the stagnation equilibrium is unstable.

A.4 The Trend-Stabilized Closure: Derivation of the Quadratic ODE

Under the trend-stabilized closure $\hat{y}^p = \theta\hat{k} + b$, the growth rate of capital productivity is:

$$\hat{b} = (\theta - 1)\hat{k} + b. \quad (34)$$

Substituting this into the acceleration identity (equation 28) generates a quadratic ODE:

$$\frac{d\hat{k}}{dt} = (\hat{k} + \delta) [(\theta - 1)\hat{k} + b]. \quad (35)$$

The equilibria are the roots of $(\hat{k} + \delta)[(\theta - 1)\hat{k} + b] = 0$:

$$\hat{k}_1^* = -\delta, \quad \hat{k}_2^* = \frac{b}{1 - \theta}. \quad (36)$$

The Jacobian $g'(\hat{k}) = 2(\theta - 1)\hat{k} + (\theta - 1)\delta + b$ evaluated at $\hat{k}_2^*$ confirms local stability for the empirically relevant case ($\theta < 1$ and $b > 0$): the interior root $b/(1 - \theta)$ acts as a strictly positive attractor, preventing stagnation. As $b \rightarrow 0$, the trend-stabilized closure collapses to the unbalanced growth attractor at $\hat{k}^* = 0$.

B Data Construction and Diagnostic Support

B.1 Series Glossary

Table 15 lists the core replication variables, their sources, and transformation rules. All series derive from Shaikh’s (2016) data companions via Real Economic Analysis.⁴

Table 15: Replication series: symbols, sources, and transformations

Symbol / Code	Variable	Source (table, line)	Transformation
$GVAcorp_t$	Corporate gross value added (corrected)	NIPA 1.14 (L1) + imputed-interest adj. (Table 7.11)	nominal
$KGCcorp_t$	Corporate gross capital stock (GPIM-adjusted)	Fixed Asset 6.1 (L9); IRS Statistics of Income; GPIM recursion	nominal
$KNCcorp_t$	Corporate net capital stock (GPIM-adjusted)	same as above	nominal
$KNCcorpbea_t$	Corporate net capital stock (BEA 2011 official)	Fixed Asset 6.1 (L9)	nominal
p_t	Common deflator	pKN index, 2005=100	$(KNCcorp/KNRcorp)*100$
y_t	Real log output	constructed	$\ln(VAcorp_t/p_t)$
k_t	Real log capital	constructed	$\ln(KGCcorp_t/p_t)$
$Profshcorp_t$	Corporate profit share (corrected)	NIPA 1.14 (L8, L1–L2) + imputed-interest adj.	ratio

Note: All BEA and NIPA data downloaded in the replication vintage. Shaikh’s original series used data downloaded in 2011 (Shaikh, 2016, Appendix 6.7, fn. 1); discrepancies attributable to subsequent BEA revisions are possible and noted where material.

Source: [Real Economic Analysis Data Tables](#)

B.2 Imputed-Interest Correction

The NIPA corporate value added figure includes imputed financial intermediation services as a cost to non-financial corporations. The corporate imputed-interest adjustment removes this component:

$$CorpImpIntAdj_t \equiv -BankMonIntPaid_t - CorpNFNetImpIntPaid_t, \quad (37)$$

where $BankMonIntPaid_t$ is NIPA Table 7.11 (lines 4+44+73 minus lines 28+52+91) and $CorpNFNetImpIntPaid_t$ is NIPA Table 7.11 (line 74 minus line 53). In 2009 the adjustment raises corporate net operating surplus by approximately 10 %. Figure 10 displays the corrected versus uncorrected series and their ratio.

B.3 Corrected Corporate Output: $GVAcorp_t$

$$GVAcorp_t \equiv GVAcorpnipa_t + CorpImpIntAdj_t, \quad (38)$$

where $GVAcorpnipa_t$ is NIPA Table 1.14 (line 1) and $CorpImpIntAdj_t$ is defined in equation (37).

⁴Replication data and construction scripts are available at Real Economic Analysis: <https://www.realecon.org/data>.

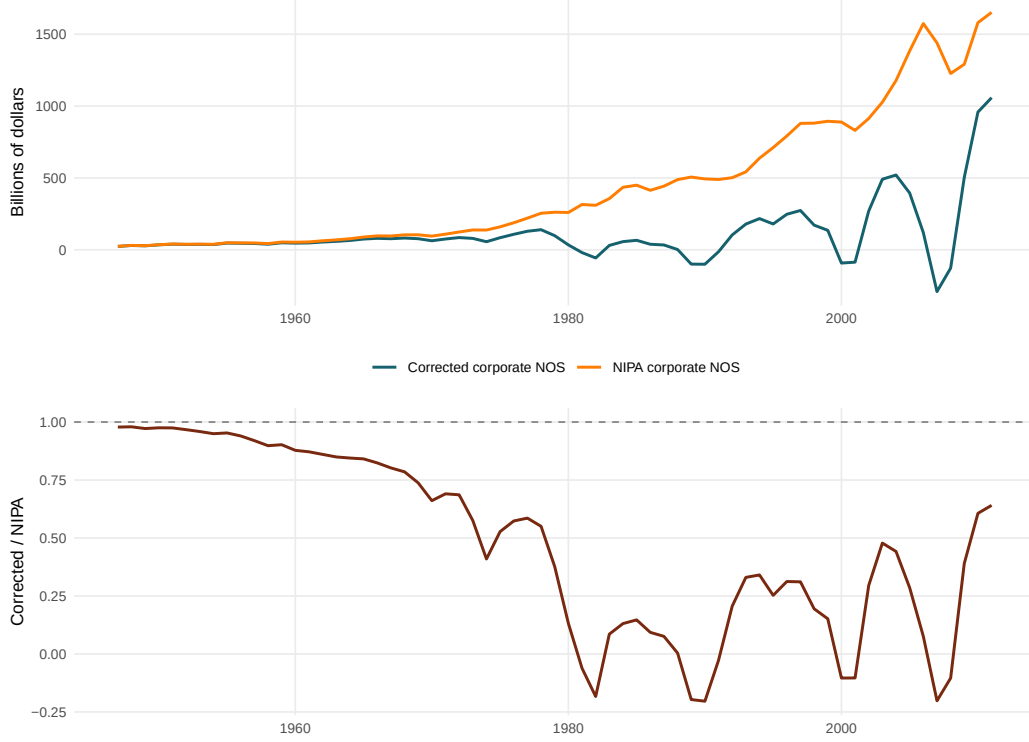


Figure 10: Imputed interest adjustment impact: corrected corporate NOS versus NIPA corporate NOS (1947–2011). Ratio shows magnitude of adjustment.

B.4 Reconstructed Corporate Capital Stock: $KGC_{corp,t}$

GPIM accumulation rule. The current-cost capital stock is accumulated as:

$$K_t = IG_t + z_t^* \cdot K_{t-1}, \quad z_t^* \equiv (1 - z_t) \frac{p_t^K}{p_{t-1}^K}, \quad (39)$$

where IG_t is nominal gross investment, z_t is the period depreciation rate, and z_t^* is the survival-and-revaluation factor that reflates the surviving stock to current replacement cost.

Three adjustments (applied before the recursion).

1. **Depletion rates.** BEA 1993 finite service lives replace the BEA 2011 infinite-lives assumption, implying higher depreciation and lower retirement rates consistent with physical exhaustion.
2. **Initial values.** BEA 1993 initial values anchored to 1925 replace the BEA 2011 starting values. The 1993 values are lower, producing faster accumulation growth and convergence to the official path by approximately 1977.
3. **Great Depression correction.** The IRS corporate book value index ([of the Census, 1975](#), Series V 115, pp. 924–926) is applied over 1925–1947, rescaling the BEA historical stock by the IRS/BEA ratio each year. The GPIM recursion resumes from 1947. Net effect: $KNC_{corp,t}$ is approximately 28 % lower in 1947 than the BEA 2011 official measure.

Inventories. IRS corporate balance sheet inventory ratios are extrapolated over 1947–1989 via a linear fit to the 1990–2011 IRS/BEA ratio, rescaled to the adjusted GPIM stocks, and added to $KGCcorp_t$. Source: IRS Statistics of Income 1926–2011; FRB Flow of Funds FL105015205.A for non-financial corporate inventories in current cost.

Accuracy. The average ratio of the GPIM approximation to the BEA current-cost stock over 1947–2005 is 99.60 %.

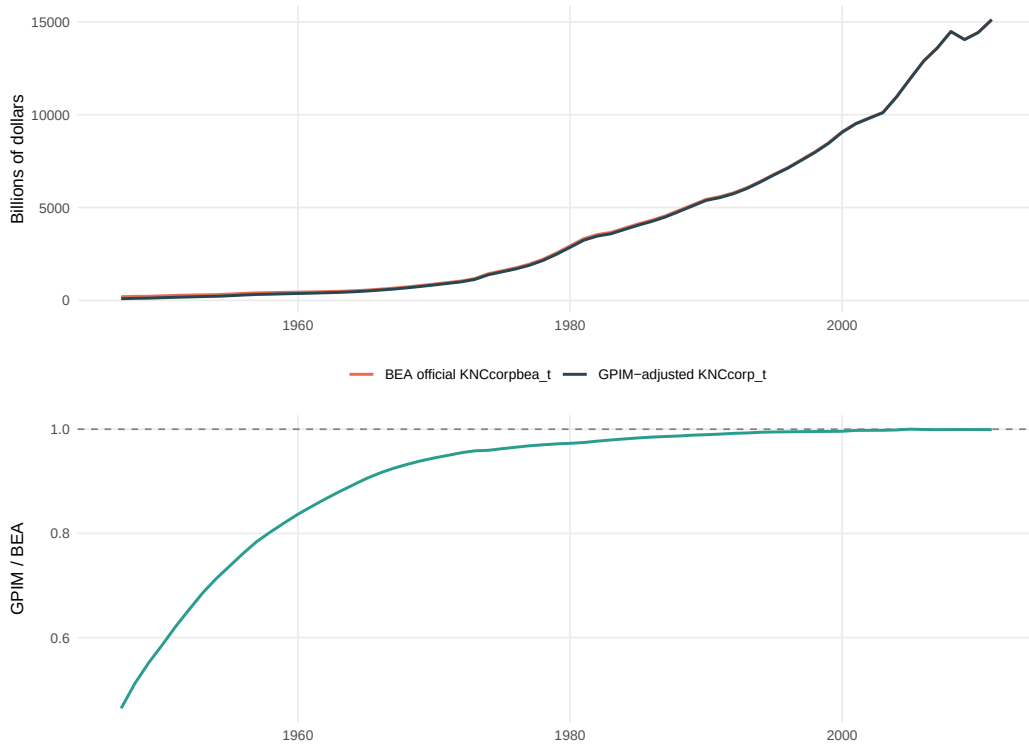


Figure 11: GPIM-adjusted net capital stock versus BEA 2011 official series (1947–2011). Ratio plot shows convergence by 1977.

B.5 Common Deflator: p_t

The replication uses the pKN implicit price deflator based at $100 = 2005$, estimated under GPIM rules. Current BEA releases employ quality-adjusted chain weights that adjust for quality vintage, breaking stock-flow consistency for accumulation accounting. The pKN index preserves stock-flow consistency, ensuring the output–capital relation reflects real productive dynamics rather than quality-adjustment artifacts (Shaikh, 2016, Appendix 6.6, Sec. II).

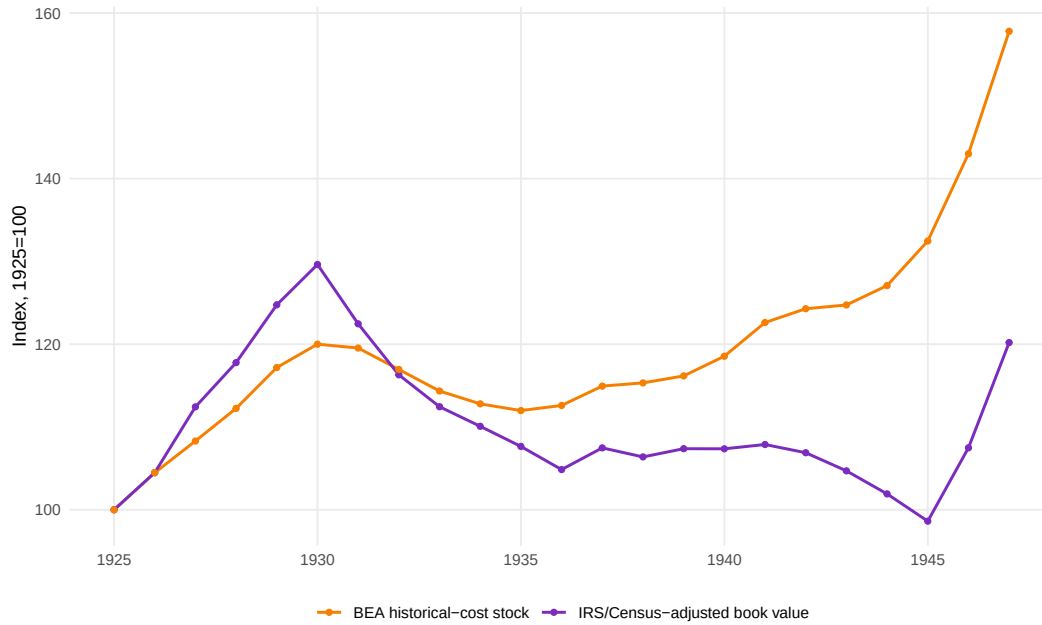


Figure 12: IRS corporate book value index versus BEA historical stock (1925–1947). Both indexed to 100 in 1925.

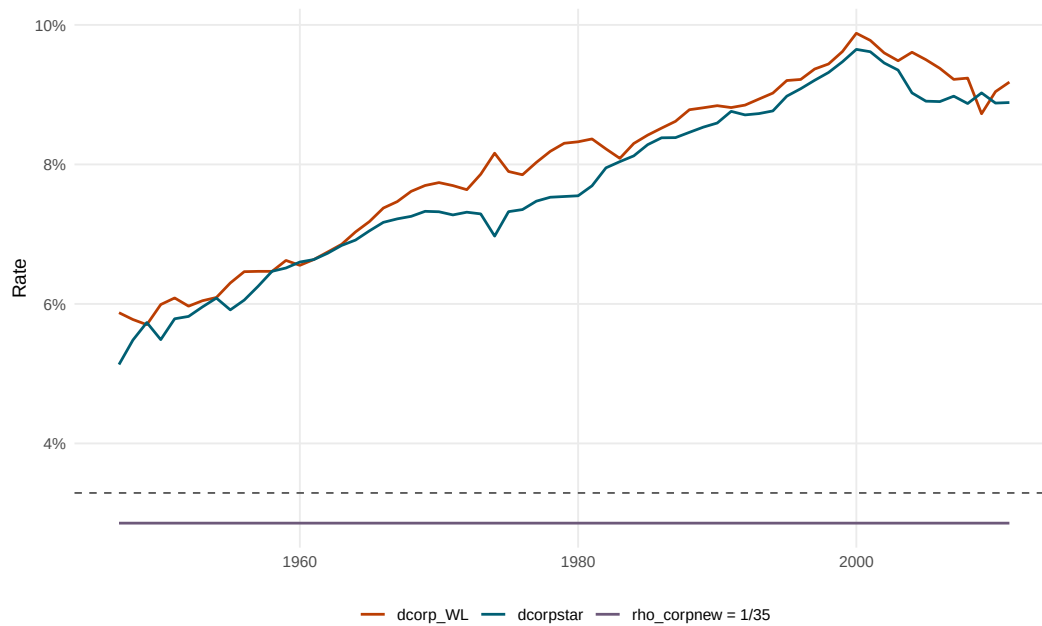


Figure 13: Depreciation and retirement rates under BEA 1993 finite service lives. Horizontal line marks critical value (3.29%).

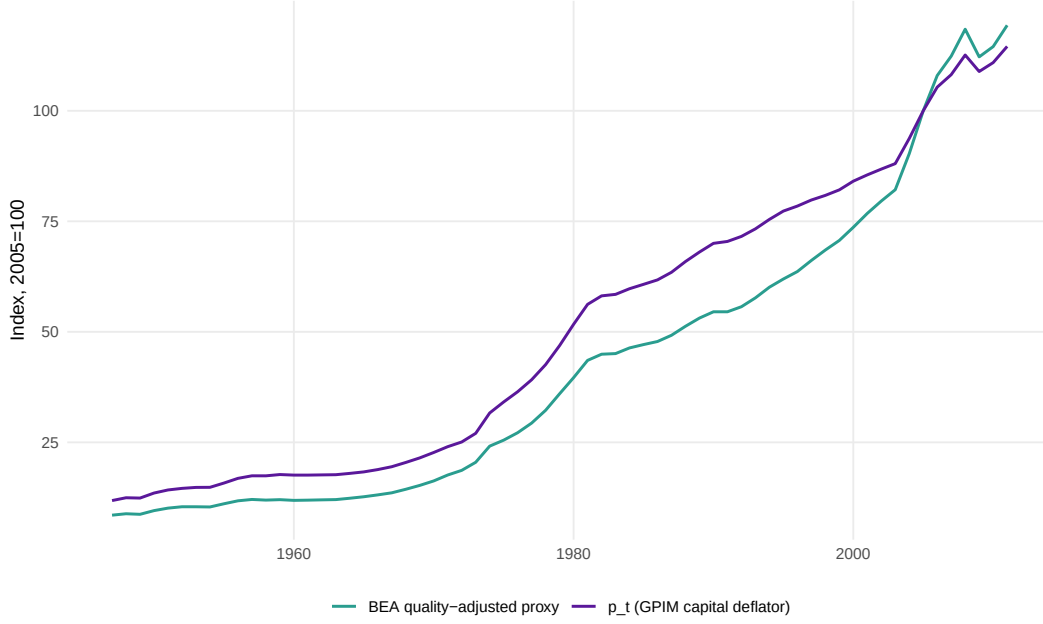


Figure 14: Common deflator index: GPIM-consistent pKN versus BEA quality-adjusted proxy (2005=100).

B.6 Corrected Corporate Profit Share: Profshcorp_t

The same imputed-interest adjustment applied to GVAcorp_t is applied to both the numerator and denominator of the profit share:

$$\text{NOScorp}_t \equiv \text{NOScorpnipa}_t + \text{CorpImpIntAdj}_t, \quad (40)$$

$$\text{VAcorp}_t \equiv \text{VAcorpnipa}_t + \text{CorpImpIntAdj}_t, \quad (41)$$

$$\text{Profshcorp}_t \equiv \frac{\text{NOScorp}_t}{\text{VAcorp}_t}, \quad (42)$$

where NOScorpnipa_t is NIPA Table 1.14 (line 8) and VAcorpnipa_t is NIPA Table 1.14 (line 1 minus line 2).

B.7 Descriptive Statistics and Unit Root Tests

Table 16 reports summary statistics for the core series. Table 17 presents unit root test results (ADF, Phillips-Perron) for y_t and k_t in levels and first differences.

Table 16: Core series summary statistics (1947–2011)

Variable	N	Mean	Median	Std Dev	Min	Max	Skewness	Kurtosis
y_t	65	14.9281	14.8653	0.5674	13.8866	15.7628	−0.1438	1.8398
k_t	65	15.9658	15.9810	0.7313	14.7979	17.1127	−0.0206	1.6872
Δy_t	64	0.0293	0.0355	0.0391	−0.0979	0.1055	−0.6331	3.4760
Δk_t	64	0.0362	0.0361	0.0078	0.0186	0.0530	−0.1127	2.4022

p_t	65	50.3486	46.8900	32.8159	11.8262	114.5539	0.3955	1.8063
D_{56}	65	0.0154	0.0000	0.1240	0.0000	1.0000	7.8750	63.0156
D_{74}	65	0.0154	0.0000	0.1240	0.0000	1.0000	7.8750	63.0156
D_{80}	65	0.0154	0.0000	0.1240	0.0000	1.0000	7.8750	63.0156

Note: Source: Real Economic Analysis (<https://www.realecon.org/data>), Shaikh (2016) data companions. Variables constructed per Table 15 of Appendix B. Dummy variables are impulse indicators with mean $1/65 \approx 0.0154$.

Table 17: Unit root tests for core series (1947–2011)

Variable	Form	Test	Specification	Statistic	Lag/BW	Critical values			Reject at 5%
						1%	5%	10%	
y_t	level	ADF	none	3.0027	1	-2.60	-1.95	-1.61	no
y_t	level	PP	intercept	-1.3280	3	-3.53	-2.91	-2.59	no
y_t	level	PP	trend	-2.1341	3	-4.11	-3.48	-3.17	no
y_t	Δ	ADF	intercept	-5.0966	1	-3.51	-2.89	-2.58	yes
y_t	Δ	PP	intercept	-5.9637	3	-3.54	-2.91	-2.59	yes
y_t	Δ	PP	trend	-5.9460	3	-4.11	-3.48	-3.17	yes
k_t	level	ADF	none	1.1552	4	-2.60	-1.95	-1.61	no
k_t	level	PP	intercept	-0.3863	3	-3.53	-2.91	-2.59	no
k_t	level	PP	trend	-1.2114	3	-4.11	-3.48	-3.17	no
k_t	Δ	ADF	intercept	-2.0670	3	-3.51	-2.89	-2.58	no
k_t	Δ	PP	intercept	-1.9027	3	-3.54	-2.91	-2.59	no
k_t	Δ	PP	trend	-1.8431	3	-4.11	-3.48	-3.17	no

Note: ADF: Augmented Dickey–Fuller test with max lag 6 and AIC selection. PP: Phillips–Perron test with Newey–West bandwidth. "none" = no deterministic terms; "intercept" = constant only; "trend" = constant + linear trend. PP "none" specification not computed. Rejection at 5% indicates stationarity under that specification.

Figures showing the log levels of output and capital, the output–capital ratio, and their annual growth rates are presented in the main text (see Section 4.2, Figures 1, 2, and 3).

B.8 ARDL Specification Search and Bounds Testing

Table 18 displays the full specification grid ($p, q = 1-6$; Cases I–V) with information criteria, diagnostic tests, and recovered long-run coefficients. Asterisks mark case-specific AIC/BIC minima.

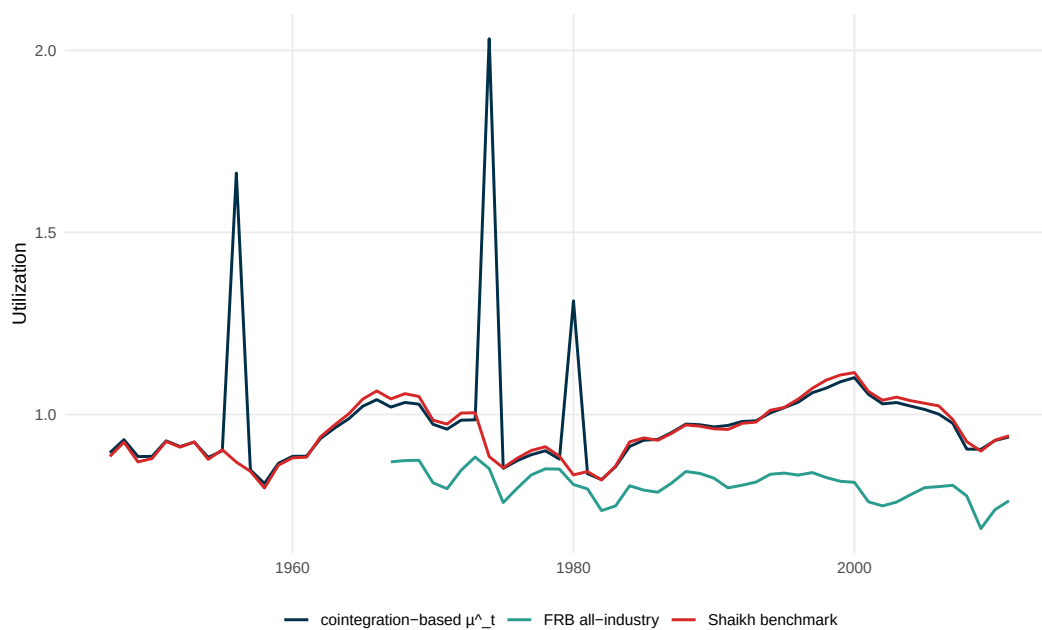


Figure 15: Capacity utilization comparison: Shaikh (cointegration-based) versus FRB survey measure, 1947–2011. Descriptive comparison only.

Table 18: ARDL specification search: information criteria and diagnostics (1947–2011)

p	q	Case	AIC	SBC	T_{eff}	$\hat{\theta}$	LM(1)	p -val	LM(4)	p -val	Best
1	1	I	−237.63	−222.52	64	0.8529	3.2749	0.070	5.0893	0.278	
1	1	II	−243.09	−225.82	64	0.6989	3.7150	0.054	6.9003	0.141	
1	1	III	−243.09	−225.82	64	0.6989	3.7150	0.054	6.9003	0.141	
1	1	IV	−243.09	−223.66	64	−0.5048	3.4486	0.063	6.5385	0.162	
1	1	V	−243.09	−223.66	64	−0.5048	3.4486	0.063	6.5385	0.162	
1	2	I	−239.33	−222.19	63	0.7273	7.0198	0.008	11.1845	0.025	
1	2	II	−242.69	−223.41	63	0.7110	7.1117	0.008	11.8692	0.018	
1	2	III	−242.69	−223.41	63	0.7110	7.1117	0.008	11.8692	0.018	
1	2	IV	−241.90	−220.47	63	−0.4718	7.0457	0.008	11.2449	0.024	
1	2	V	−241.90	−220.47	63	−0.4718	7.0457	0.008	11.2449	0.024	
1	3	I	−244.02	−224.88	62	0.8967	11.3499	0.001	14.1599	0.007	
1	3	II	−249.33	−228.05	62	0.6695	10.3605	0.001	12.7048	0.013	
1	3	III	−249.33	−228.05	62	0.6695	10.3605	0.001	12.7048	0.013	
1	3	IV	−249.62	−226.22	62	−0.8749	10.3560	0.001	11.9612	0.018	*
1	3	V	−249.62	−226.22	62	−0.8749	10.3560	0.001	11.9612	0.018	*
2	3	I	−250.91	−229.63	62	0.9301	5.4621	0.019	9.3503	0.053	
2	3	II	−262.79	−239.40	62	0.7023	0.5641	0.453	3.5633	0.468	*
2	3	III	−262.79	−239.40	62	0.7023	0.5641	0.453	3.5633	0.468	*
2	3	IV	−264.33	−238.81	62	−0.4836	0.2849	0.594	2.6102	0.625	*
2	3	V	−264.33	−238.81	62	−0.4836	0.2849	0.594	2.6102	0.625	*
4	5	I	−253.33	−224.01	60	0.7899	0.0461	0.830	7.6617	0.105	*
4	5	II	−257.75	−226.33	60	0.7102	0.2628	0.608	6.6363	0.156	
4	5	III	−257.75	−226.33	60	0.7102	0.2628	0.608	6.6363	0.156	
4	5	IV	−257.11	−223.60	60	−0.4117	0.2120	0.645	6.5141	0.164	
4	5	V	−257.11	−223.60	60	−0.4117	0.2120	0.645	6.5141	0.164	

Note: ARDL(p, q) specifications estimated with impulse dummies D_{56}, D_{74}, D_{80} . AIC/SBC: Akaike/Schwarz information criteria (lower = better). T_{eff} : effective sample size after lag construction. $\hat{\theta}$: long-run output–capital coefficient recovered from ECM parameters. LM(1)/LM(4): Breusch–Godfrey serial correlation tests (null = no autocorrelation). Asterisk (*) marks best specification by AIC/SBC within each deterministic case. Full grid ($p, q = 1$ –6; Cases I–V) available in replication code.

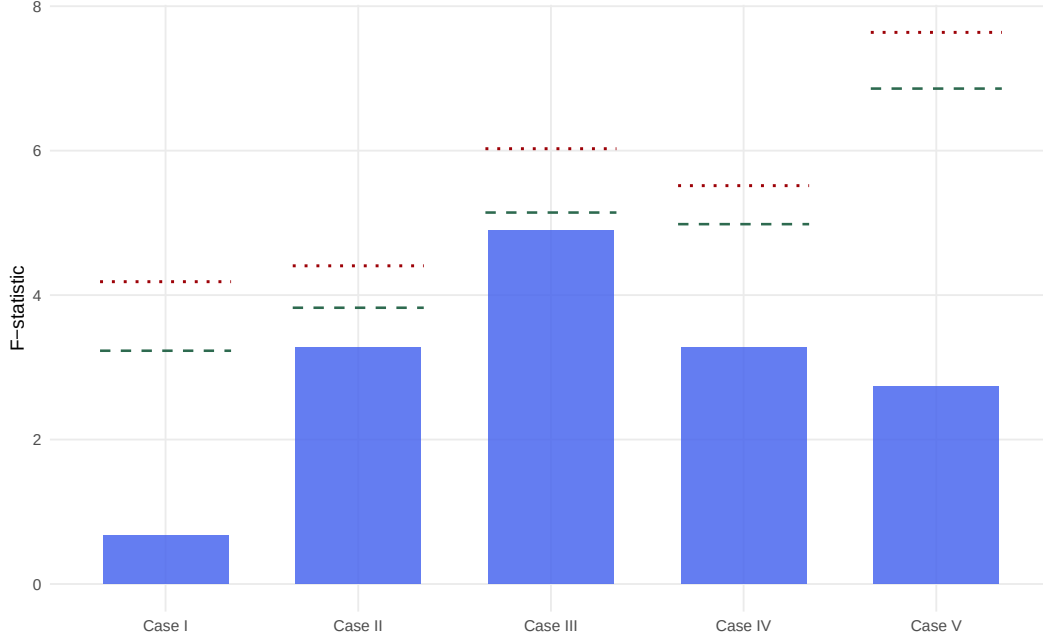


Figure 16: F -statistic across deterministic Cases I–V. Horizontal lines mark lower ($I(0)$) and upper ($I(1)$) critical bounds at 5% significance.

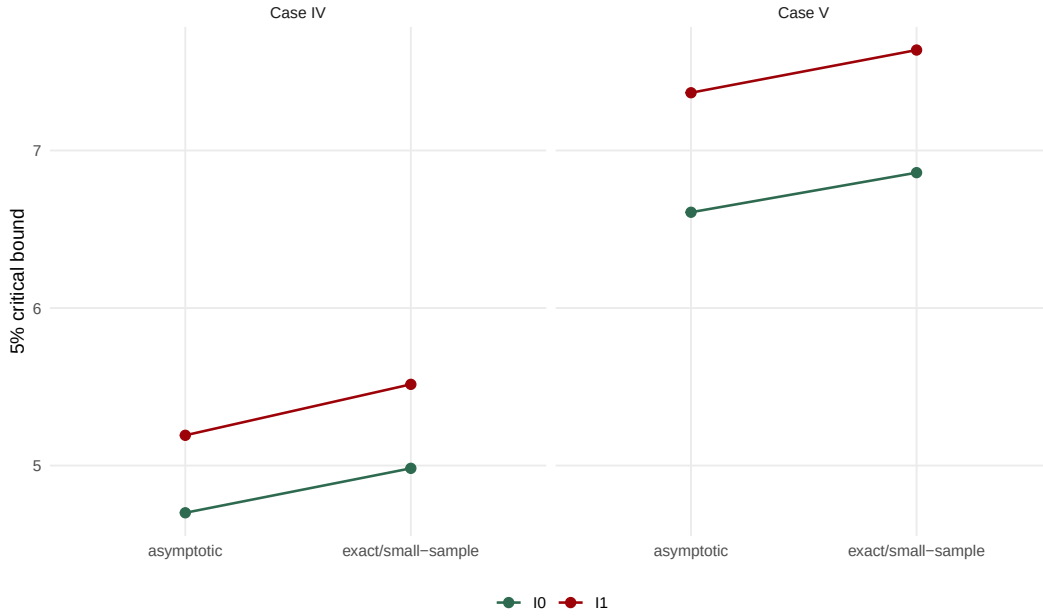


Figure 17: Exact-sample versus asymptotic critical bounds for Cases IV and V ($T = 65$). Exact bounds computed via stochastic simulation (Natsiopoulos and Tzeremes, 2022).

B.9 Dummy Variable Diagnostics

Figure 20 displays squared residuals from the baseline ARDL specification with impulse dummies D_{56} , D_{74} , D_{80} overlaid. The dummies absorb outlier-driven residual spikes identified via squared-error diagnostics (Patterson, 2000).

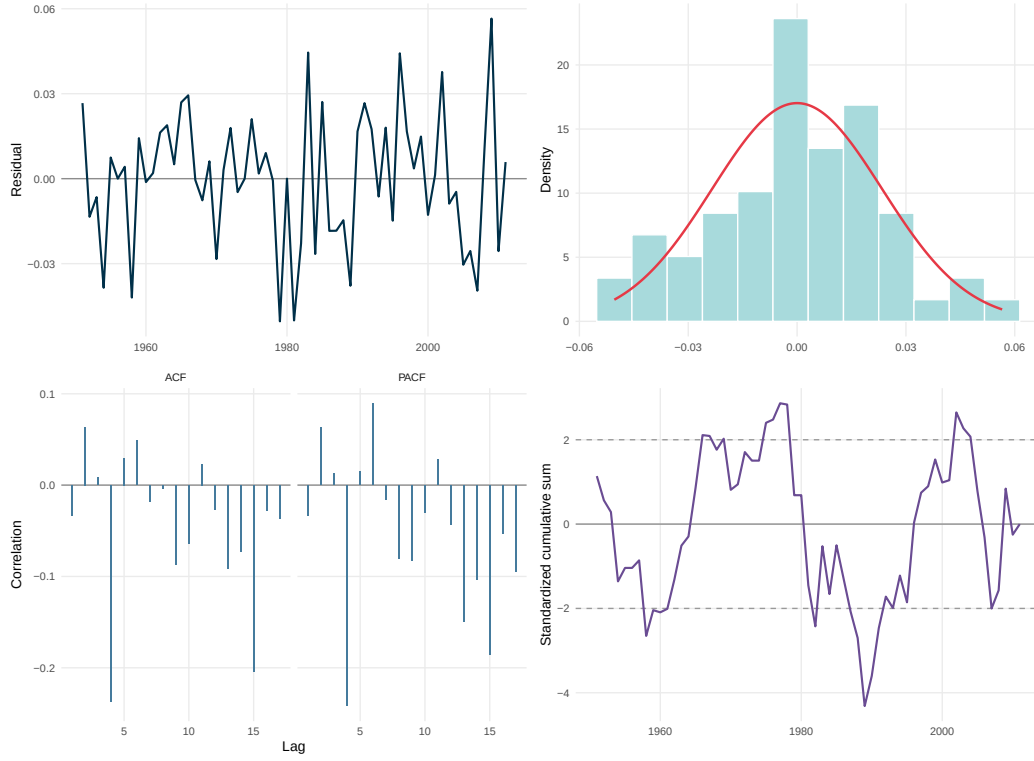


Figure 18: Residual diagnostics from baseline ARDL(2,4): (a) residuals over time, (b) density with normal curve, (c) ACF/PACF, (d) CUSUM test.

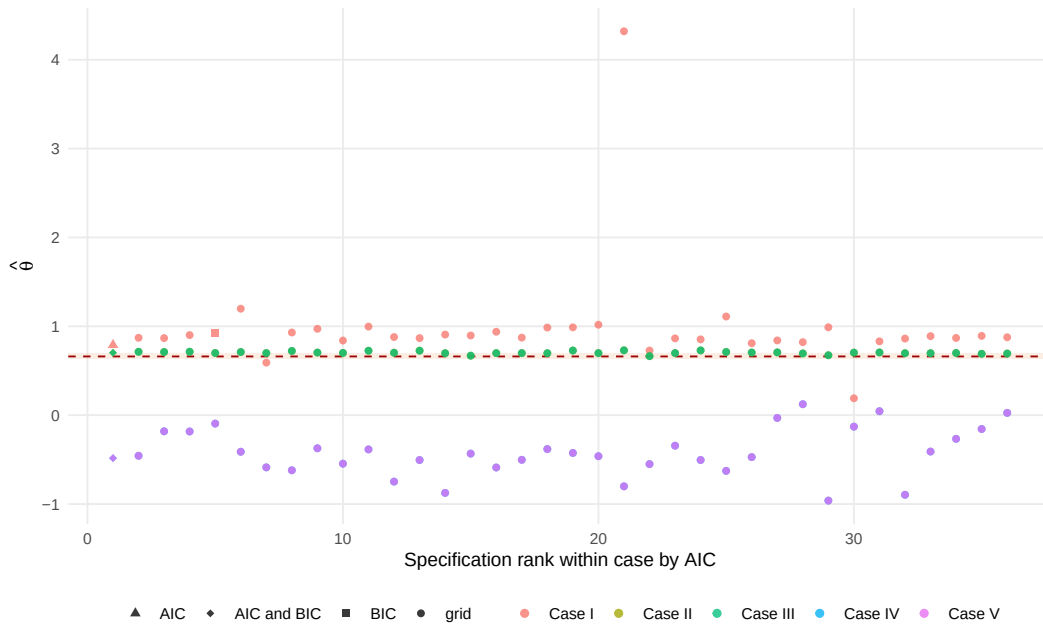


Figure 19: Long-run coefficient $\hat{\theta}$ estimates ranked by AIC across specification grid. Horizontal band marks Shaikh's $0.6609 \pm 5\%$.

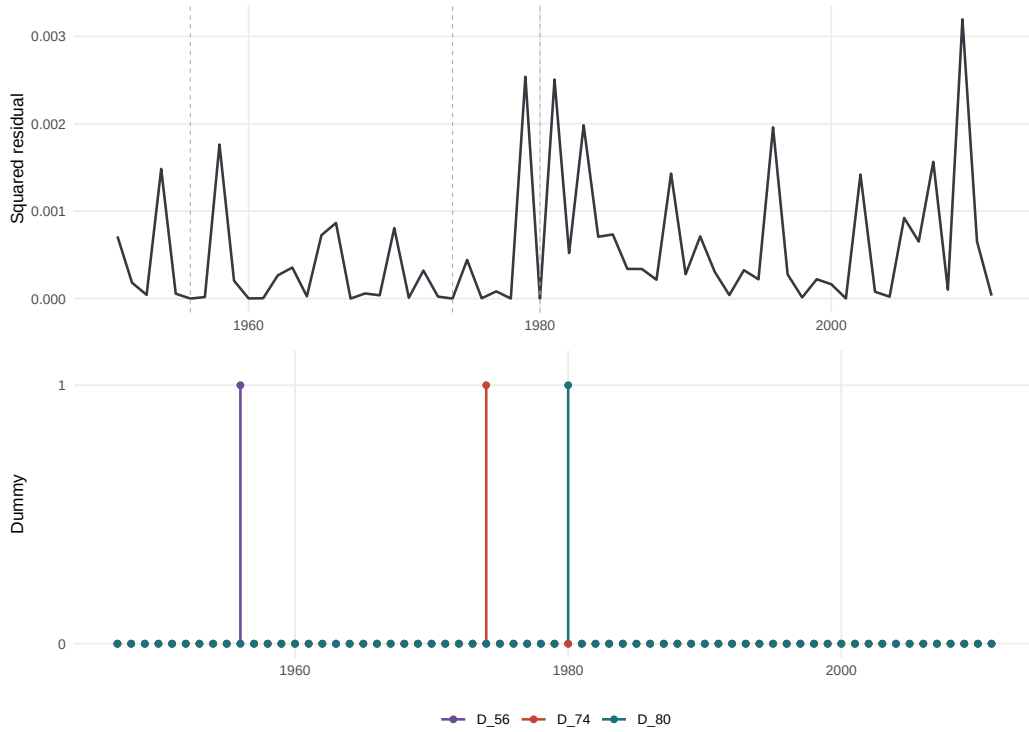


Figure 20: Squared residuals from baseline ARDL with impulse dummies D_{56} , D_{74} , D_{80} (1947–2011). Dummies absorb outlier-driven spikes.

B.10 Series Construction Protocol

Figure 21 visualizes the pipeline from raw data to fitted capacity utilization ($\hat{\mu}_t$). The flowchart traces corrections, deflation, GPIM recursion, ARDL estimation, and residualization.

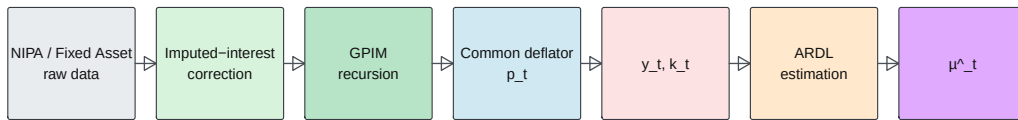


Figure 21: Series construction protocol: raw data $\rightarrow$ corrected output/GPIM capital $\rightarrow$ common deflator $\rightarrow$ ARDL estimation $\rightarrow \hat{\mu}_t$.